Funded by
the European Union
MSCA-PF 101149931

Genomics of Gene Expression Lab

CONSEJO SUPERIOR DE INVESTIGACIONES CIENTÍFICAS

Long-reads transcriptomics

Máster en Bioinformática
Universitat de València 2026

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Marie Skłodowska-Curie grant agreement No 101149931

Problems reconstructing isoforms with short-reads

Advantages:
•Higher accuracy
•Higher throughput
•Cost-effective

Disadvantages:
•Repetitive sequences
•Structural variants
•Isoform resolution –
Alternative splicing

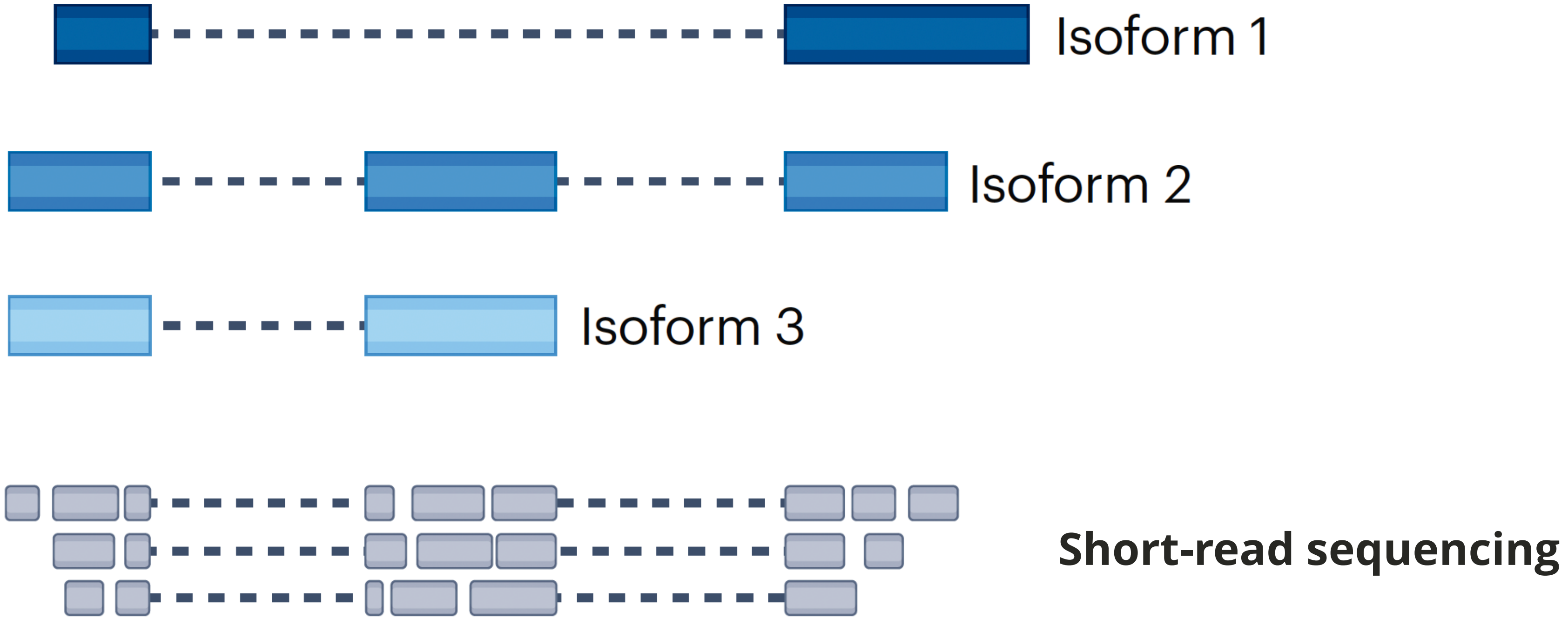
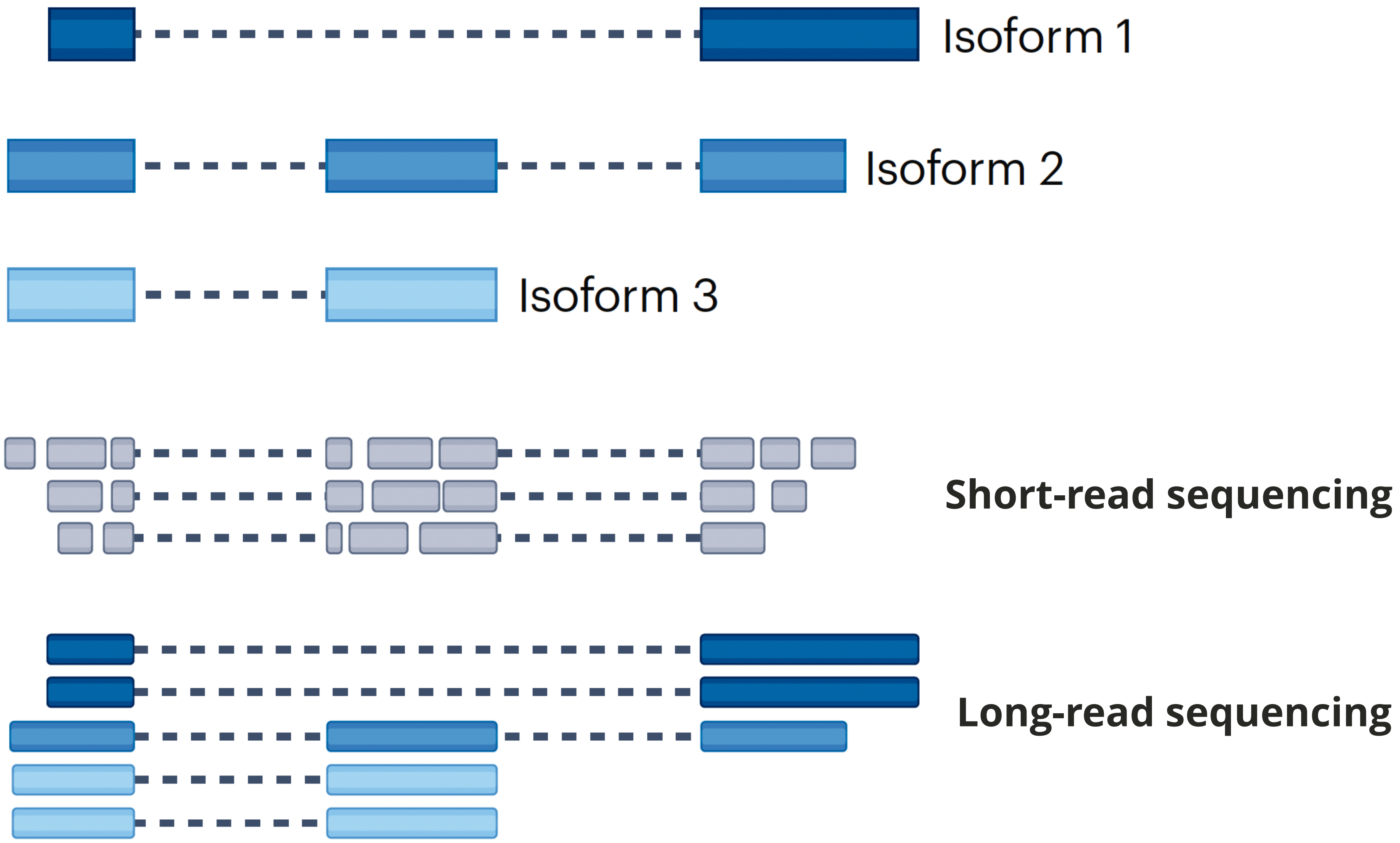


Figure source: Monzó, Liu, Conesa. 2025. 10.1038/s41576-025-00828-z

Long-read transcriptomics captures full-length transcripts

Advantages:
- Single-molecule
- Isoform resolution –
  Alternative splicing
- Assembly simplification

Disadvantages:
- Lower accuracy
- Lower throughput
- Higher costs
- Bioinformatic challenges



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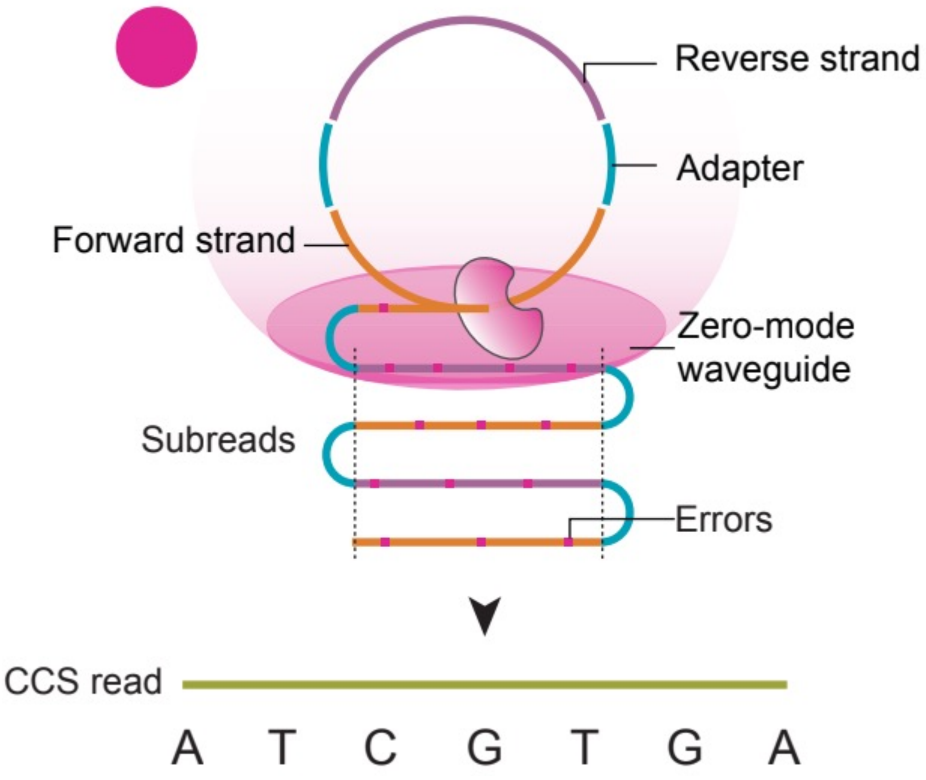
Introduction to 3rd generation long-read sequencing

PacBio



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Introduction to 3rd generation long-read sequencing

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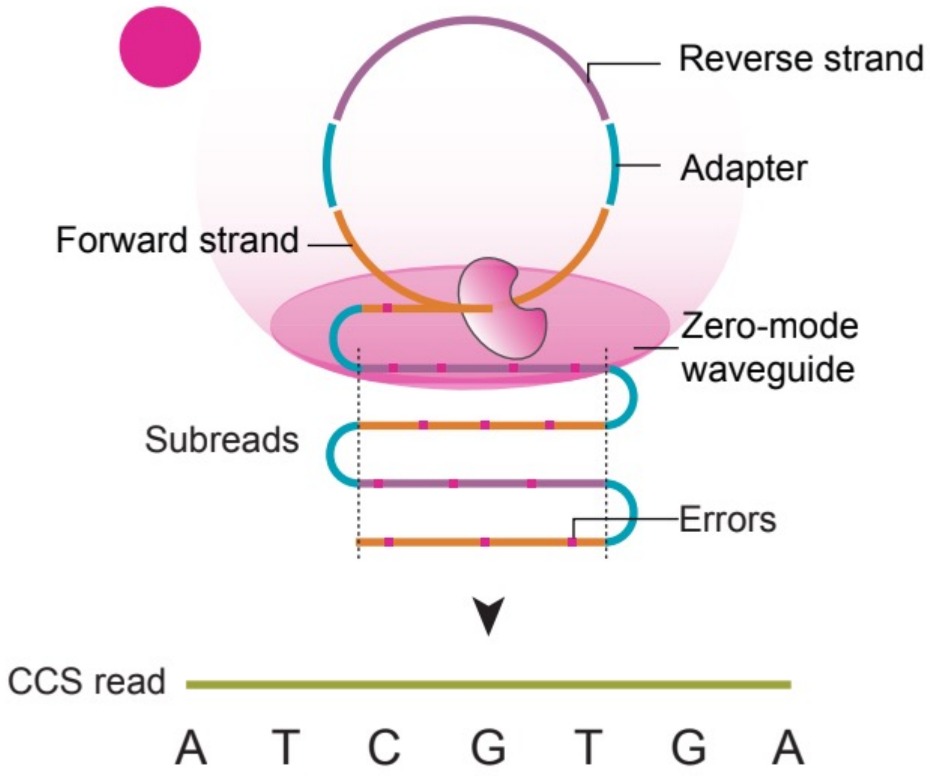

Oxford NANOPORE Technologies

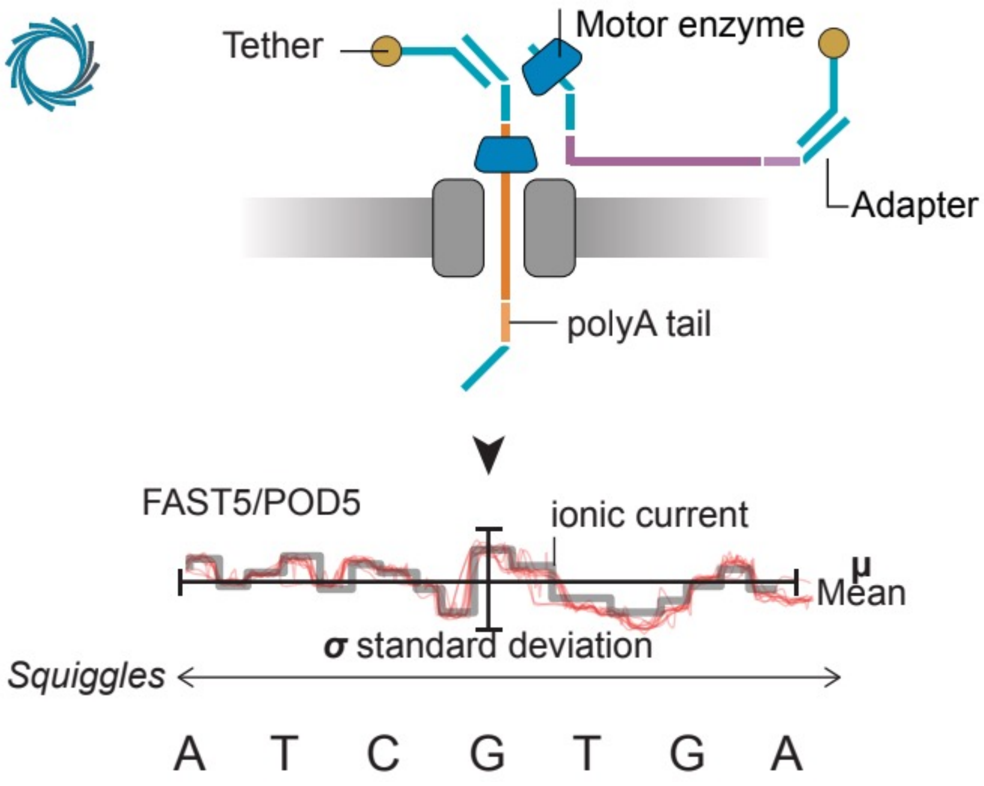

Figure source: Monzó, Liu, Conesa. 2025. 10.1038/s41576-025-00828-z

Choosing a sequencing platform

Similarities between PacBio and ONT:

- Throughput (high number of reads per run)
- Capacity to sequence full-length transcripts

Technology	Platform	Median read length (kb)	Median throughput per run (10^6 transcript reads)
PacBio	cDNA+IsoSeq+Sequel II	~2.1	~2.6
	cDNA+Kinnex+Sequel II	~1.7	~40
	cDNA+Kinnex+Revio	~1.7	~100
ONT	cDNA+MinION (R10.4)	~0.939	~20
	cDNA+PromethION (R9.4)	~1	~130
	dRNA+MinION (R9.4)	~0.8	~1.1
	dRNA+PromethION (R9.4)	~0.6	~20

Figure source: Monzó, Liu, Conesa. 2025. 10.1038/s41576-025-00828-z

Choosing a sequencing platform

PacBio

- **Higher read accuracy**
- Commercial protocols have broad read length distribution

- Can't sequence dRNA or direct modifications

Oxford NANOPORE Technologies

- Can sequence cDNA or dRNA
- Can simultaneously sequence dRNA and epitranscriptomic modifications

- Specific protocols can sequence ultra-long reads. **But standard protocols are biased towards short reads**
- Lower read accuracy
- Difficulty differentiating modifications

lrRNA-seq reveals MANY novel transcripts… some are errors?

nature

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nature > articles > article

Article | Published: 03 August 2022

Transcriptome variation in human tissues revealed by long-read sequencing

Dafni A. Glinos, Garrett Garborcauskas, Paul Hoffman, Nava Ehsan, Lihua Jiang, Alper Gokden, Xiaoguang Dai, François Aguet, Kathleen L. Brown, Kiran Garimella, Tera Bowers, Maura Costello, Kristin Ardlie, Ruiqi Jian, Nathan R. Tucker, Patrick T. Ellinor, Eoghan D. Harrington, Hua Tang, Michael Snyder, Sissel Juul, Pejman Mohammadi, Daniel G. MacArthur, Tuuli Lappalainen & Beryl B. Cummings

Nature 608, 353–359 (2022) | Cite this article

Abstract

Regulation of transcript structure generates transcript diversity and plays an important role in human disease[1,2,3,4,5,6,7]. The advent of long-read sequencing technologies offers the opportunity to study the role of genetic variation in transcript structure[8,9,10,11,12,13,14,15,16]. In this Article, we present a large human long-read RNA-seq dataset using the Oxford Nanopore Technologies platform from 88 samples from Genotype-Tissue Expression (GTEx) tissues and cell lines, complementing the GTEx resource. We identified just over 70,000 novel transcripts for annotated genes, and validated the protein expression of 10% of novel transcripts. We

Genome Biology

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Research | Open Access | Published: 07 July 2022

A high-resolution single-molecule sequencing-based Arabidopsis transcriptome using novel methods of Iso-seq analysis

Runxuan Zhang, Richard Kuo, Max Coulter, Cristiane P. G. Calixto, Juan Carlos Entizne, Wenbin Guo, Yamile Marquez, Linda Milne, Stefan Riegler, Akihiro Matsui, Maho Tanaka, Sarah Harvey, Yubang Gao, Theresa Wießner-Kroh, Alejandro Paniagua, Martin Crespi, Katherine Denby, Asa ben Hur, Enamul Huq, Michael Jantsch, Artur Jarmolowski, Tino Koester, Sascha Laubinger, Qingshun Quinn Li, ... John W. S. Brown + Show authors

Genome Biology 23, Article number: 149 (2022) | Cite this article

Abstract

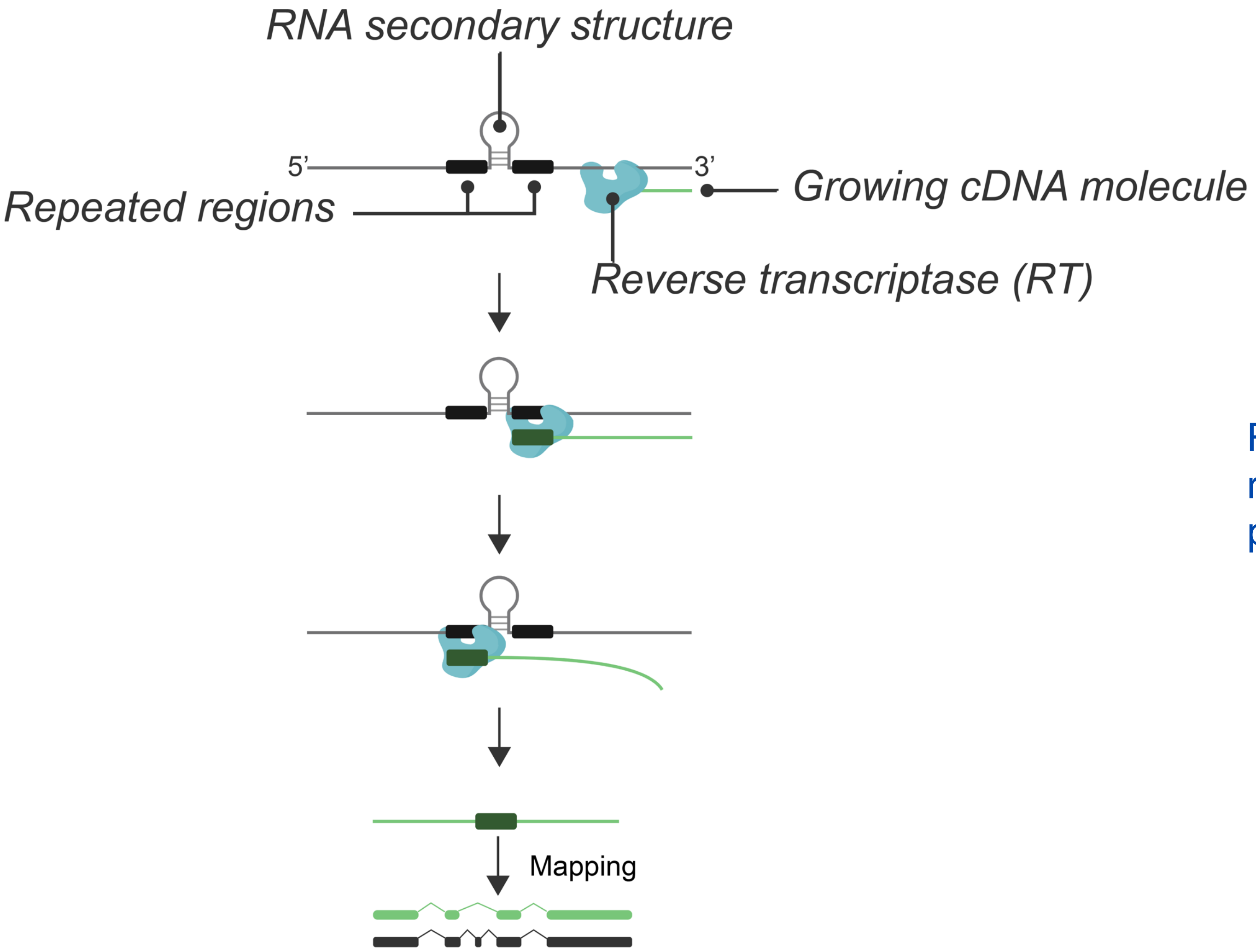
Background

Accurate and comprehensive annotation of transcript sequences is essential for transcript quantification and differential gene and transcript expression analysis. Single-molecule long-read sequencing technologies provide improved integrity of transcript structures including alternative splicing, and transcription start and polyadenylation sites. However, accuracy is significantly affected by sequencing errors, mRNA degradation, or incomplete cDNA synthesis.

Results

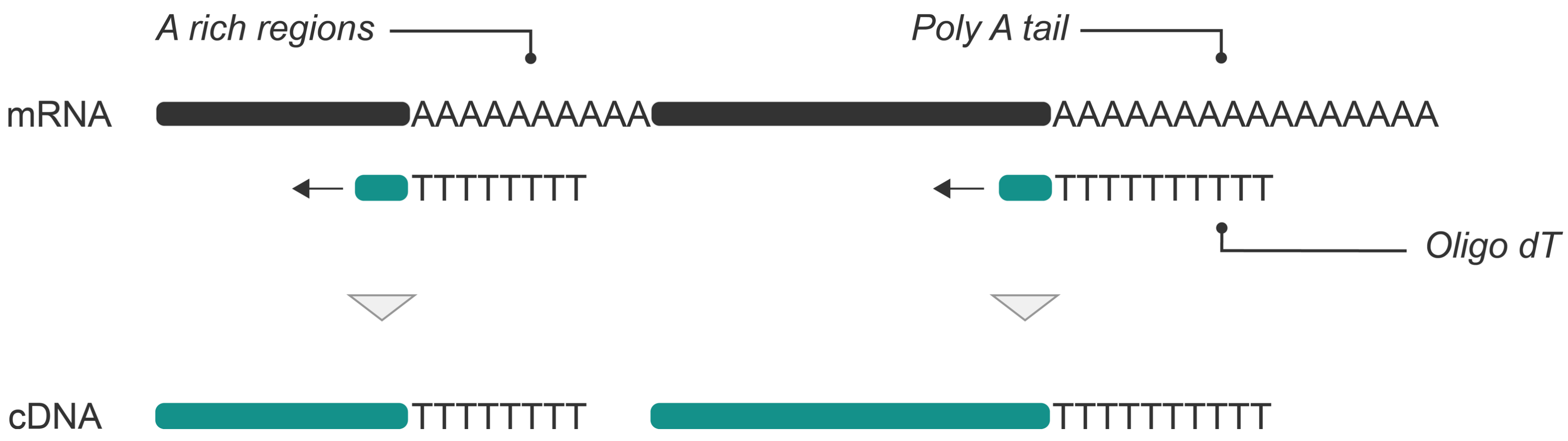
We present a new and comprehensive *Arabidopsis thaliana* Reference Transcript Dataset 3 (AtRTD3). AtRTD3 contains over 169,000 transcripts—twice that of the best current Arabidopsis transcriptome and including over 1500 novel genes. Seventy-eight percent of

Library prep errors: Reverse transcriptase template switching (RT-switching)



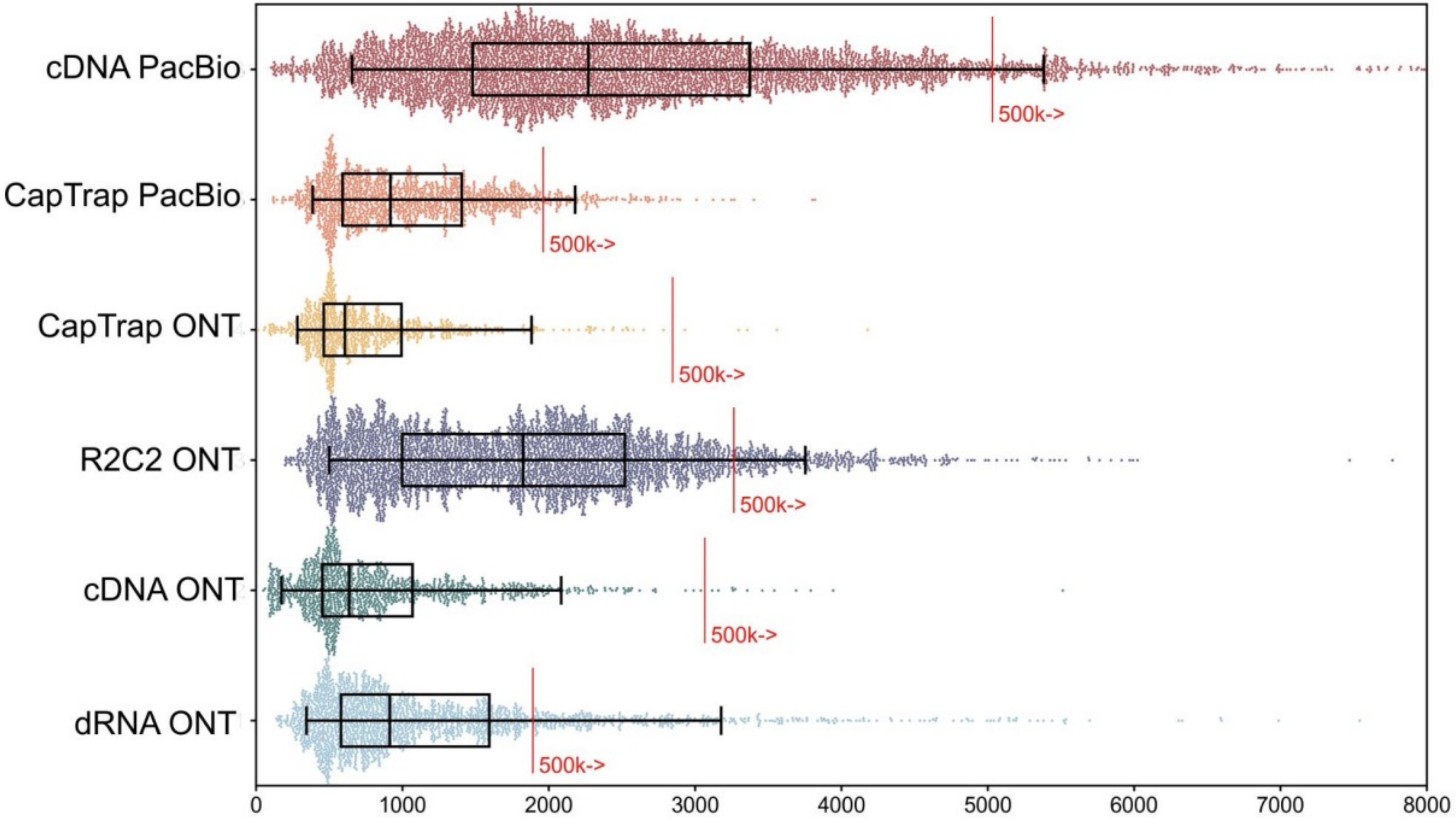
RT enzyme can ”jump” between templates at repeated regions, creating chimeric cDNA products that confound transcript analysis

Library prep errors: intra-priming



Oligo-dT primers can bind to internal A-rich regions, causing truncated cDNA synthesis

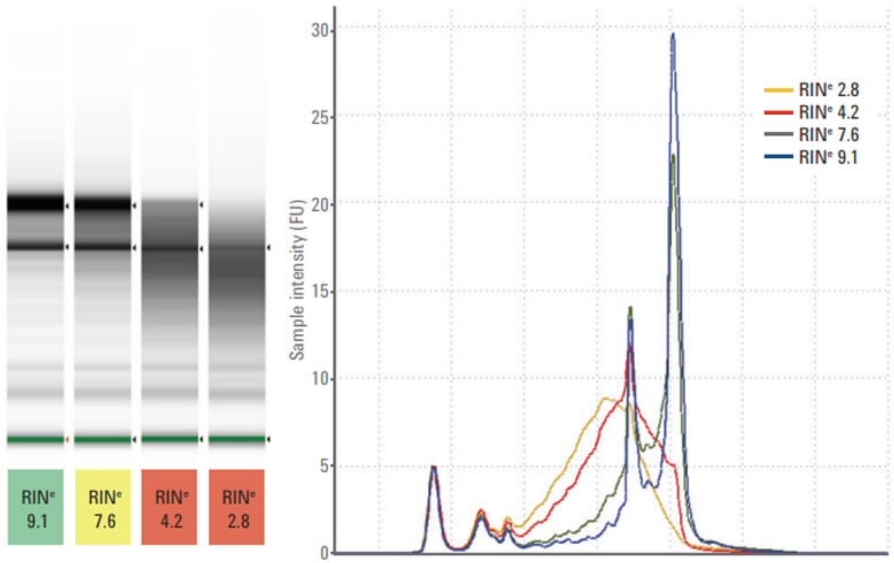
Impact of sequencing platform and library preparation choice

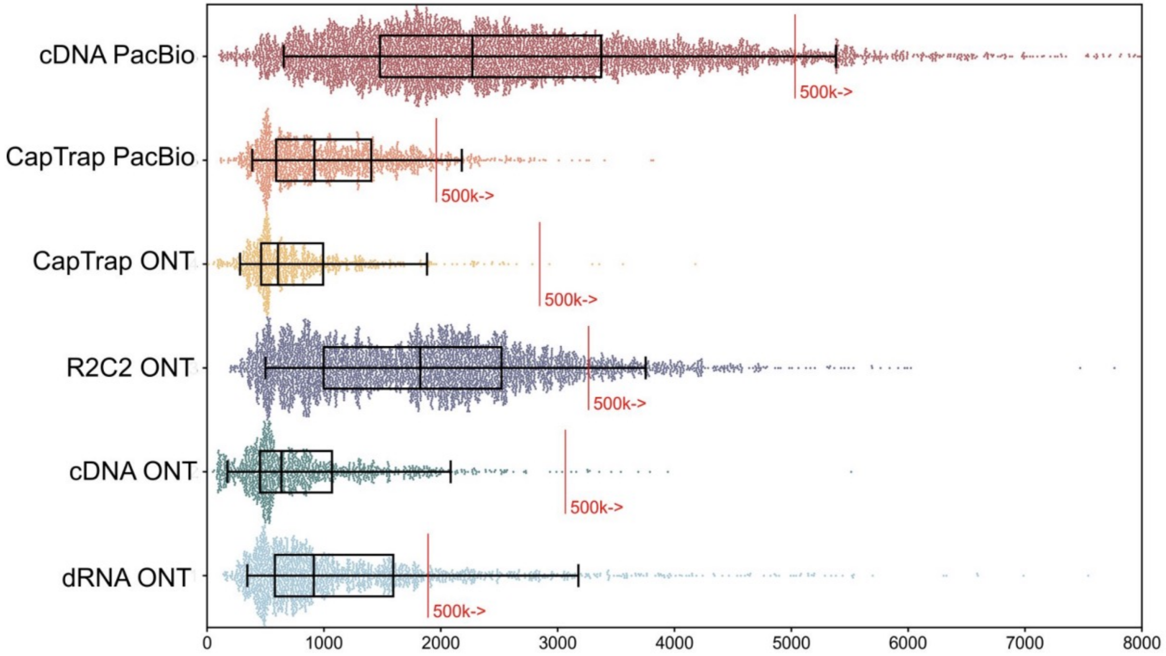


WARNING!! Combination of sequencing platform and library preparation find **different transcripts**

Figure source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

RNA quality

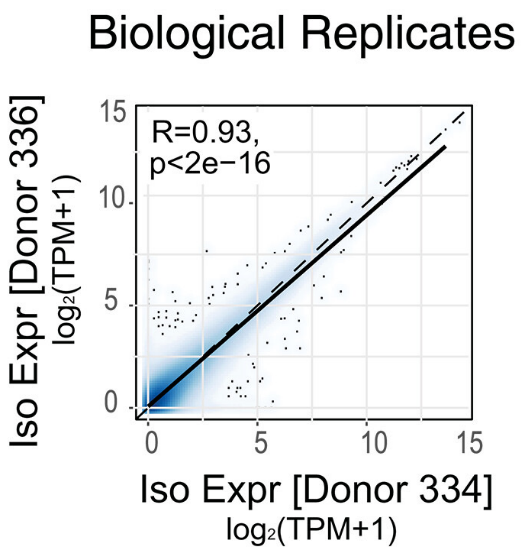




Checking the RNA quality is not only important to get good quality full-length transcripts, but also to see the distribution of transcripts that we have in the transcriptome of our sample

Figure source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

Replication and multiplexing



Replication is necessary for statistical power and to reduce the inclusion of spurious transcripts in downstream analyses.

High expression correlation between replicates

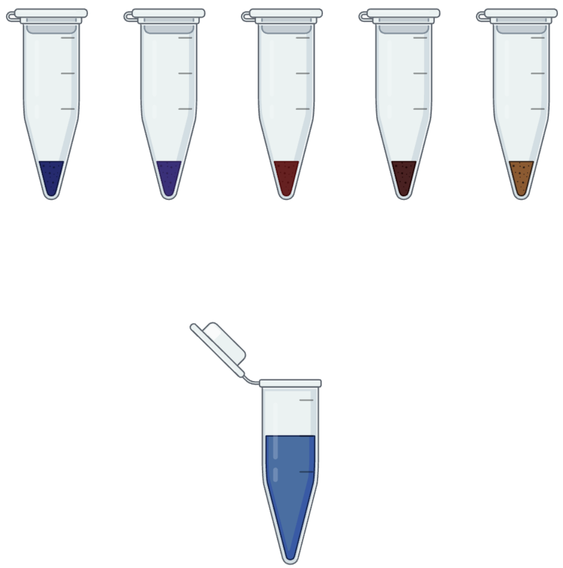


Figure source: Patowary, Zhang et al. 2024. 10.1126/science.adh7688

Replication and multiplexing

PacBio

bc = 4 different barcodes for library prep

cx = 12 different barcodes for cDNA synthesis

Possible to have 48 samples in one run

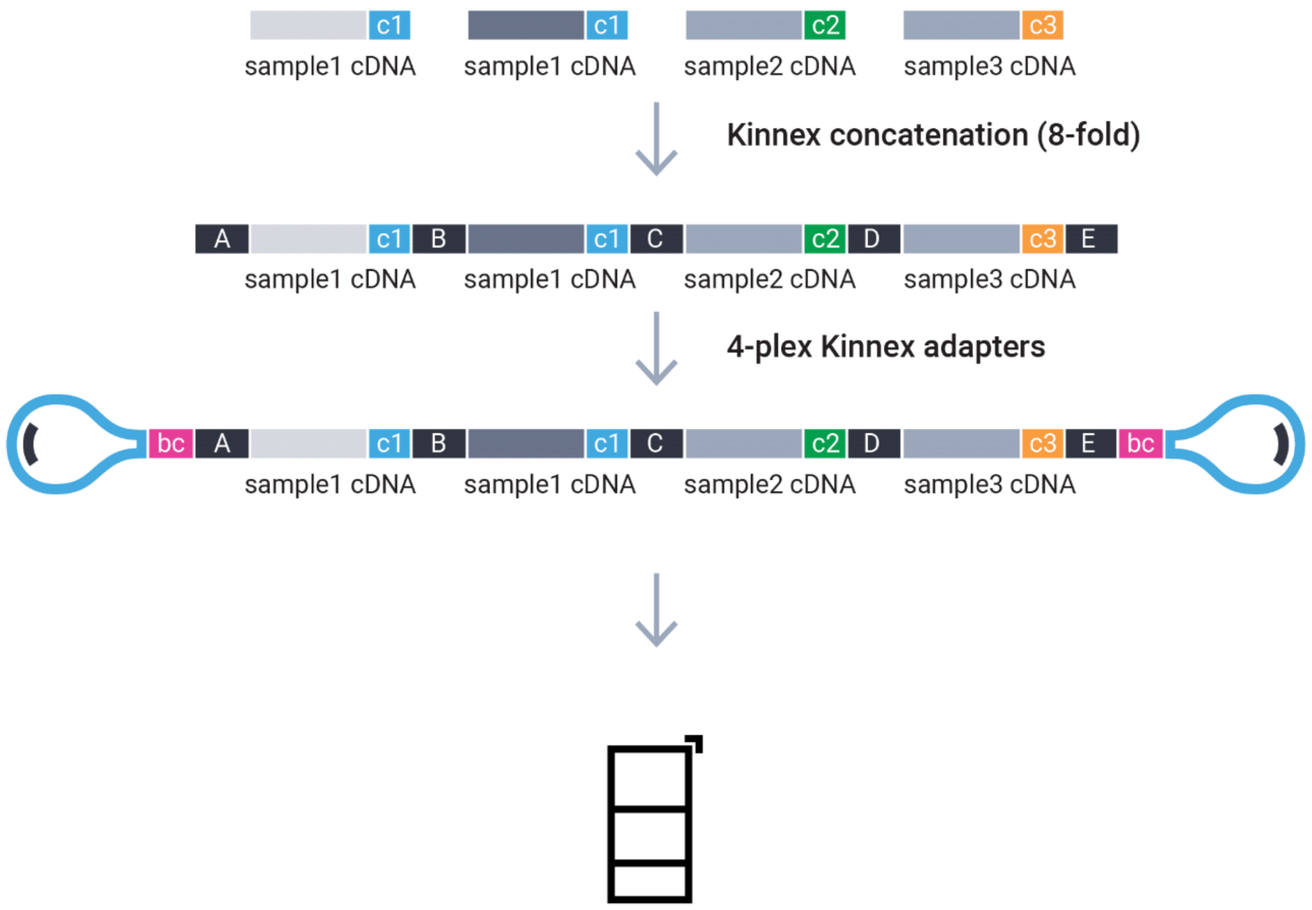
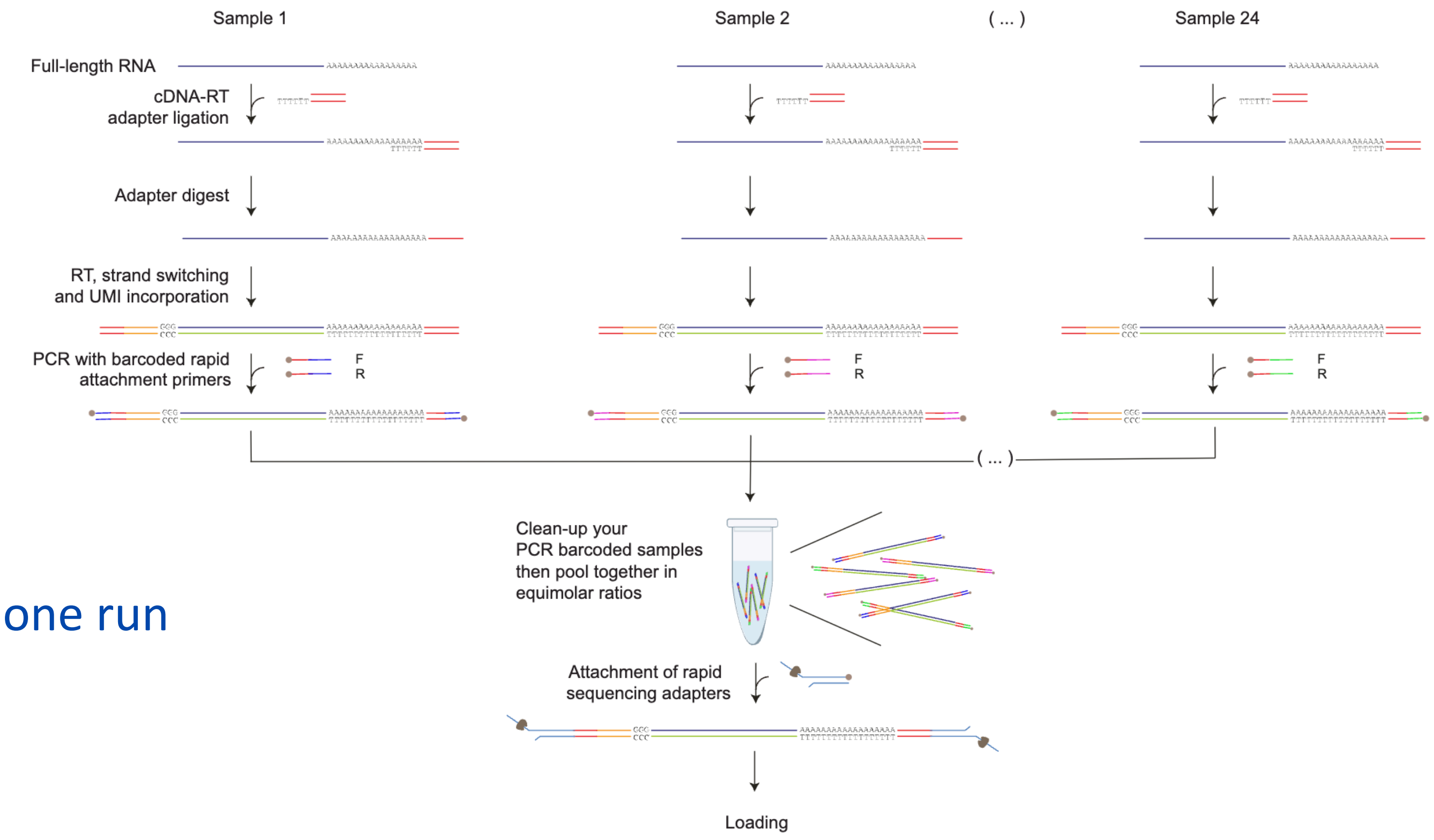


Figure source: https://www.pacb.com/multiplexing/

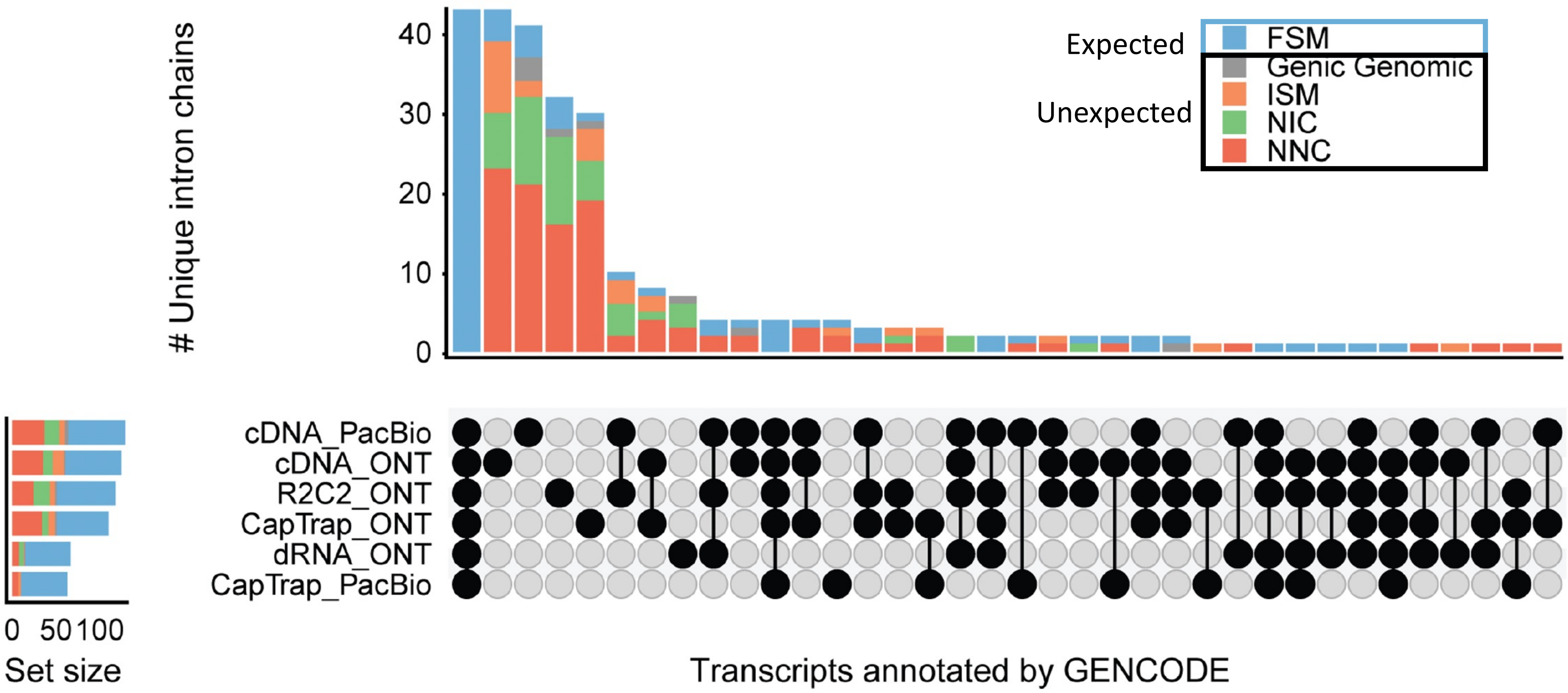
Replication and multiplexing

For cDNA!!

Possible to have 24 samples in one run



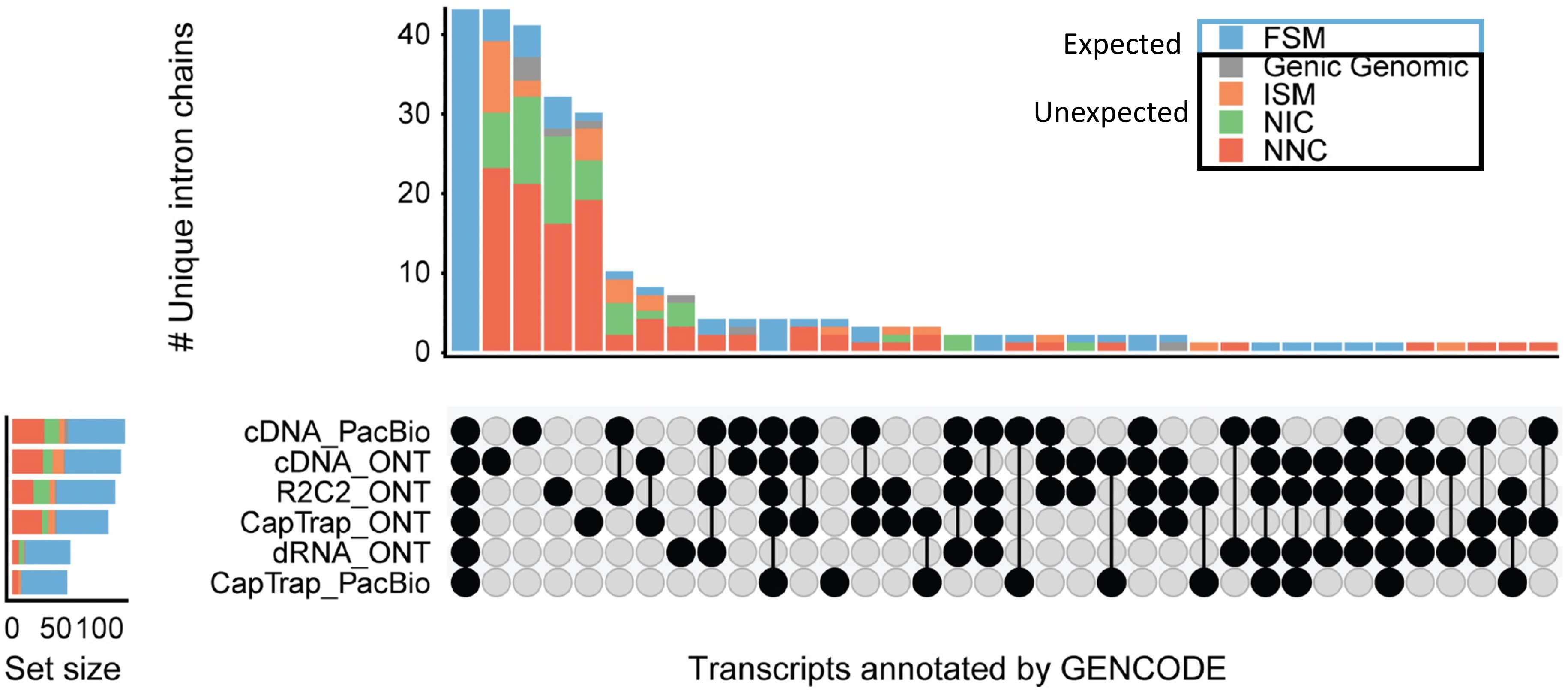
Impact of replication



Bimodal distribution, **most** transcripts are present in **all** samples **or** are **sample-specific**, few are found in two or more samples

Figure source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

Manual annotation and PCR indicate MANY novel transcripts are real



- 50 loci selected by GENCODE
- Manual annotation for each experimental dataset done INDEPENDENTLY
- Most transcripts were NOVEL and found in only ONE experimental dataset
- Validated by PCR

Figure source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

Ground truth spike-ins

Synthetic RNA molecules that mimic: isoforms, abundance, and transcript length

SIRVs and sequins

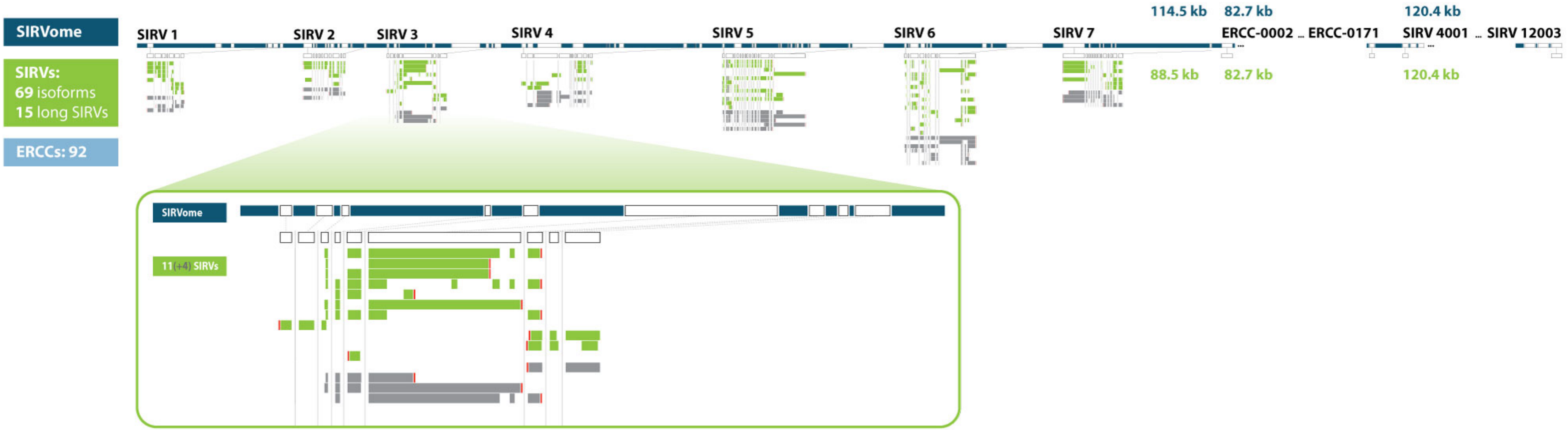
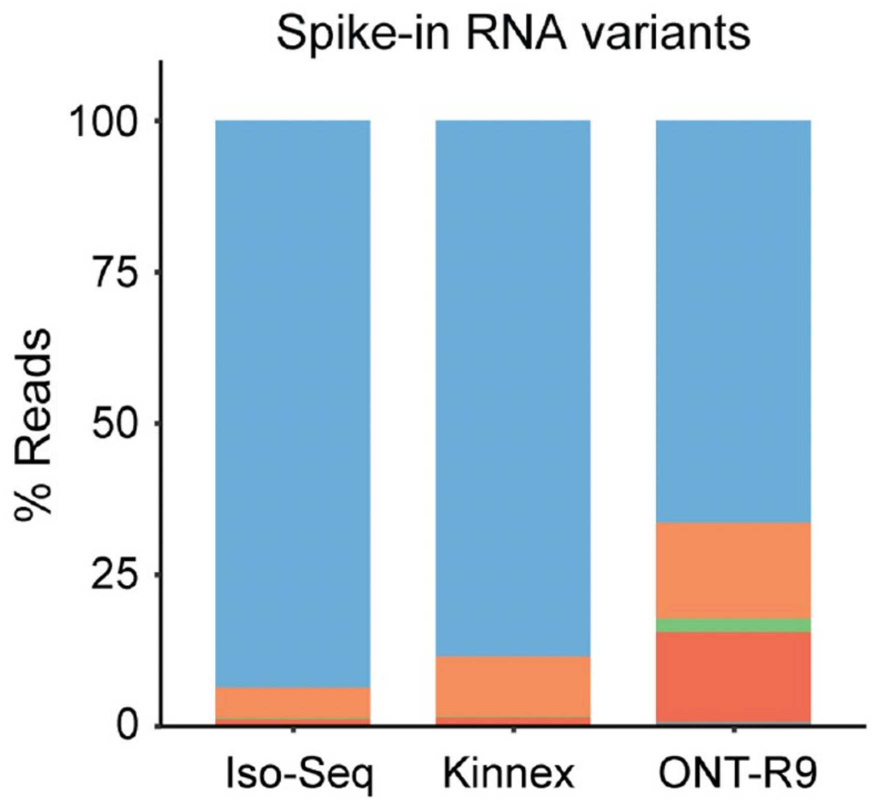
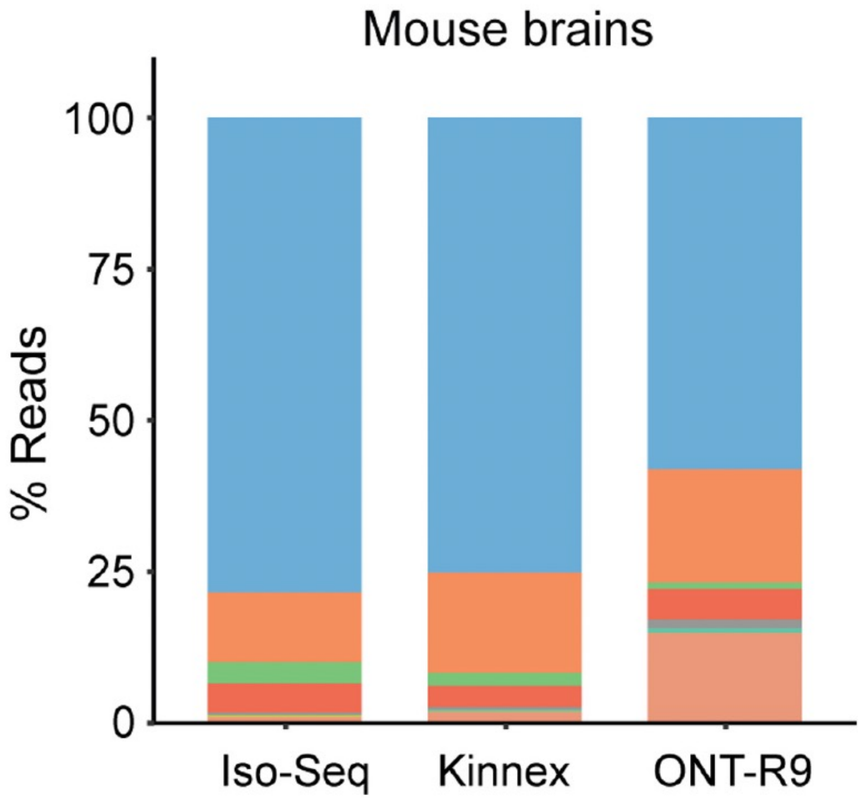


Figure source: https://www.lexogen.com/sirvs/design/

Ground truth spike-ins





- Non-blue reads in **spike-ins** represent the technical **noise level of the technology**
- Non-blue reads in the **mouse brains** are a combination of **technical and biological noise**

Figure source: Monzó, Frankish, Conesa. 2025. 10.1101/gr.279865.124

Ground truth spike-ins

Depending on the platform/library, we have 5 to 30% of problems

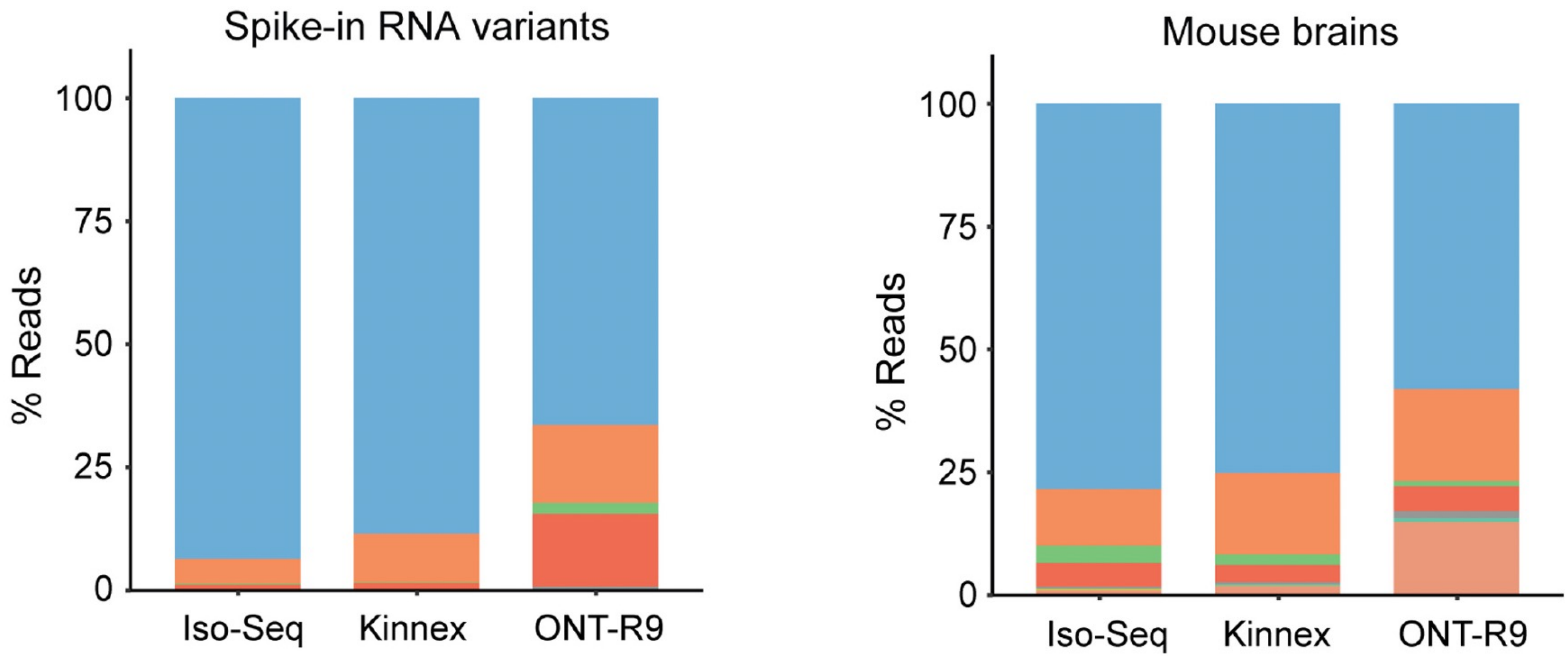


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Figure source: Monzó, Frankish, Conesa. 2025. 10.1101/gr.279865.124

Summary of challenges

- Decent depth of sequencing but nothing close to Illumina
- RT switching creates chimeric reads
- Intra-priming truncates cDNA synthesis
- Different library preparations capture different transcripts
- Technology and library prep have an impact on length and depth of sequencing
- RNA degradation… are these alternative transcription start sites?
- Replication helps, but also raises questions… few barcodes mean few replicates
- Many novel transcripts are technical artifacts… many others are real biology
- **Most** transcripts are present in **all** samples **or** are **sample-specific**

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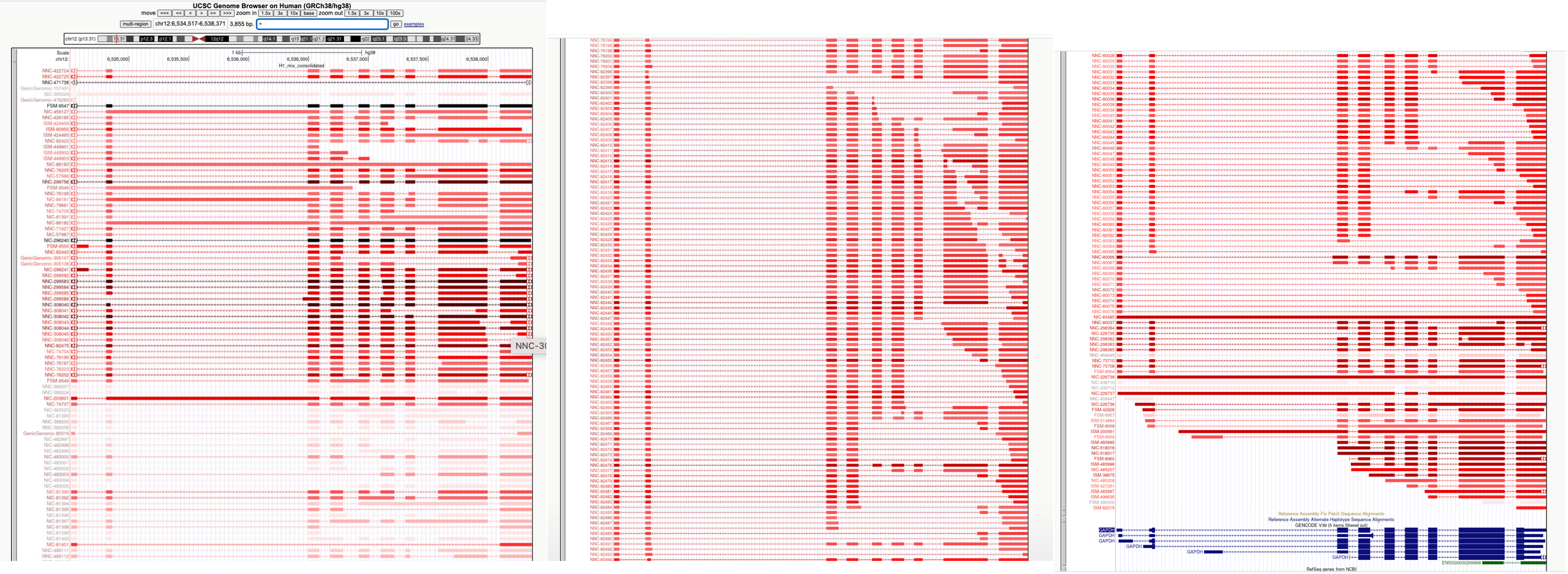
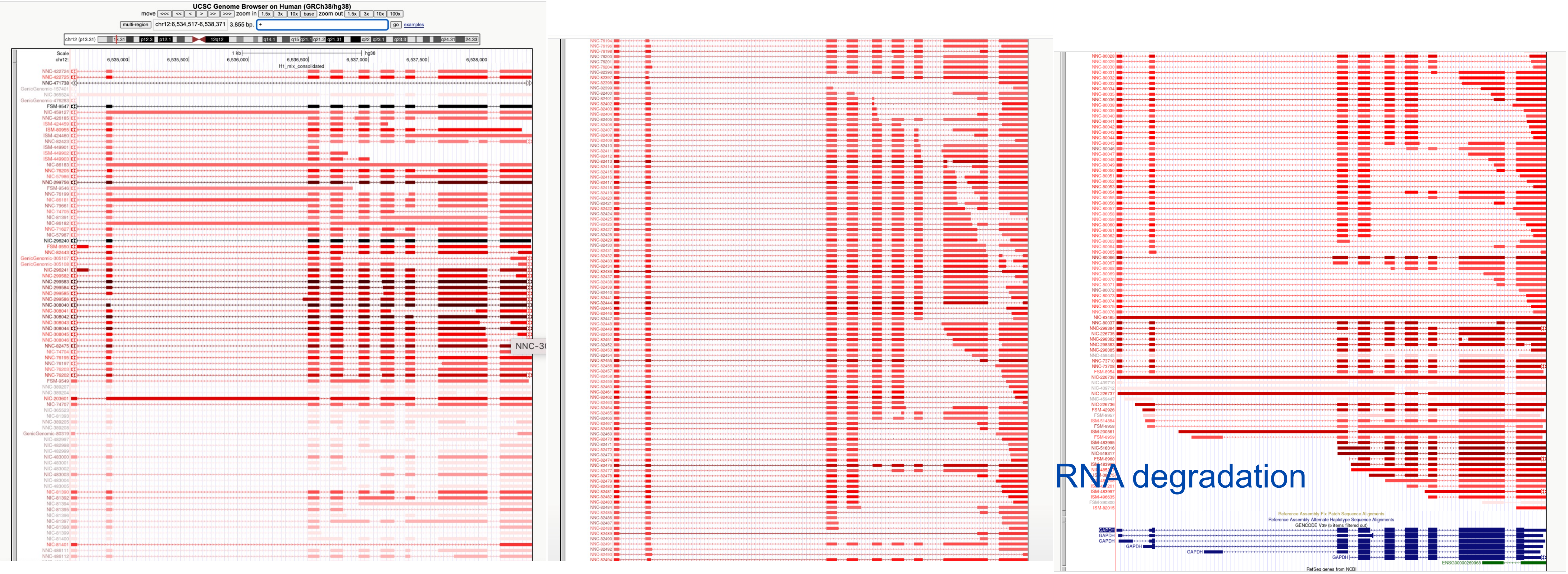
Biology is very variable, let’s try to extract it with bioinformatics

A glance on the actual challenge – Transcript models for GADPH gene



Transcripts follow different splicing patterns

A glance on the actual challenge – Transcript models for GADPH gene



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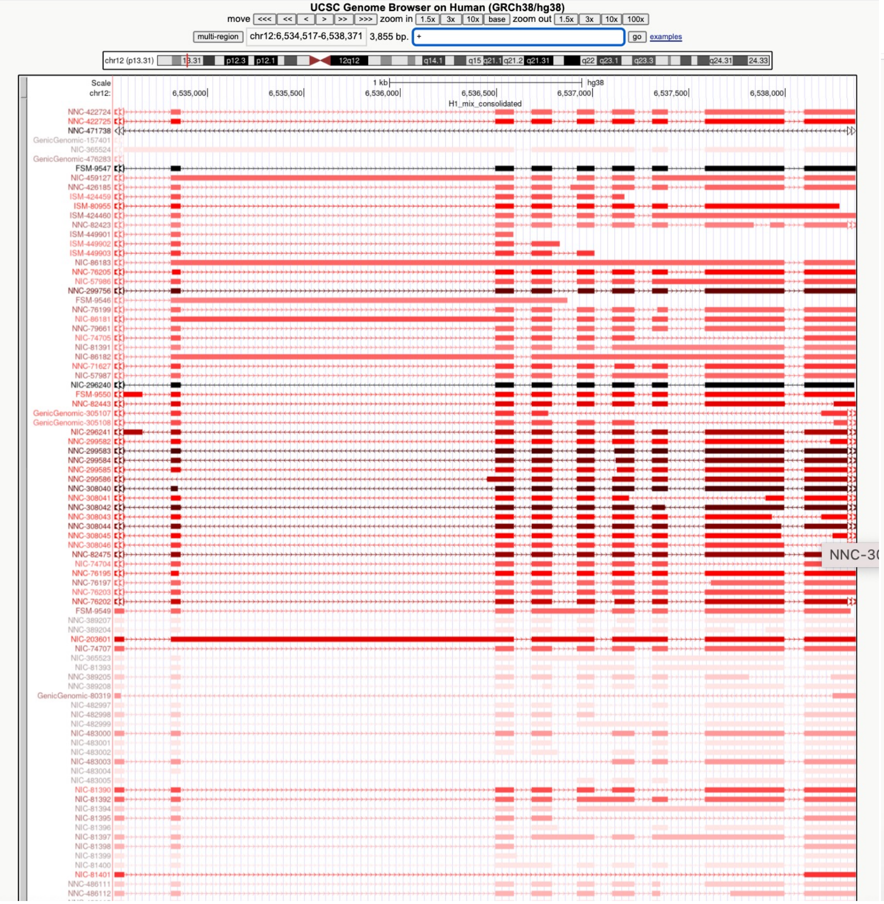
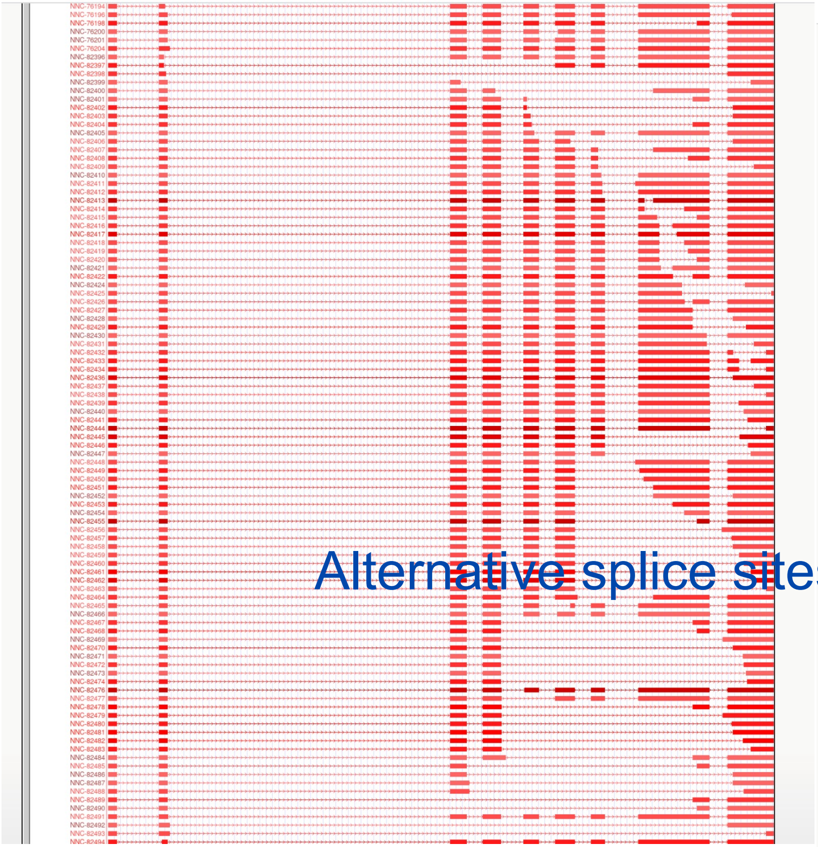
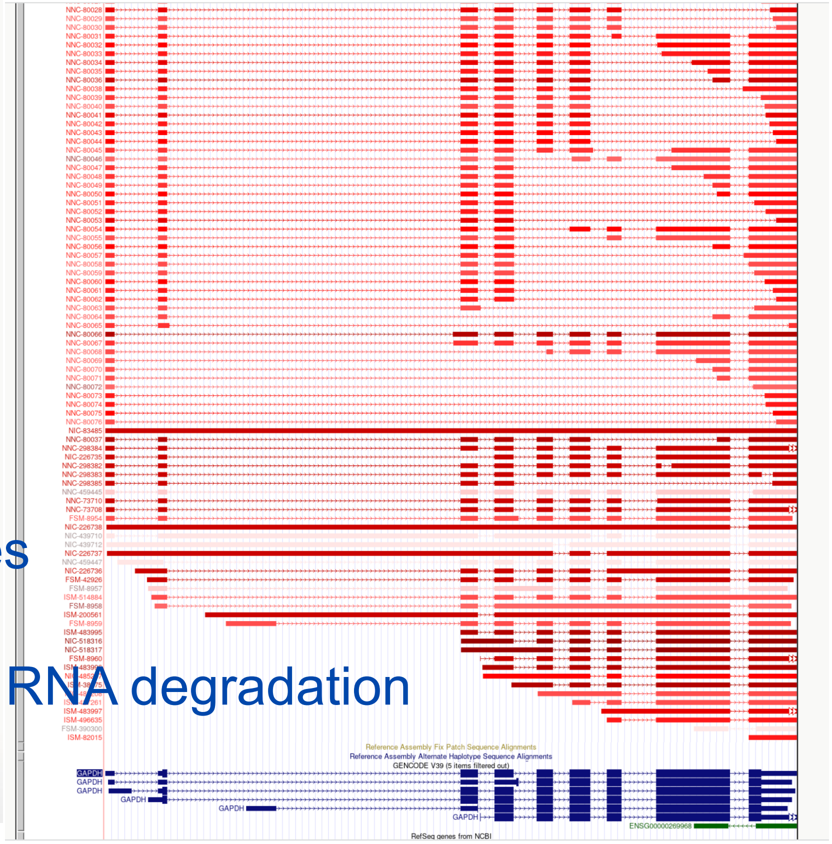




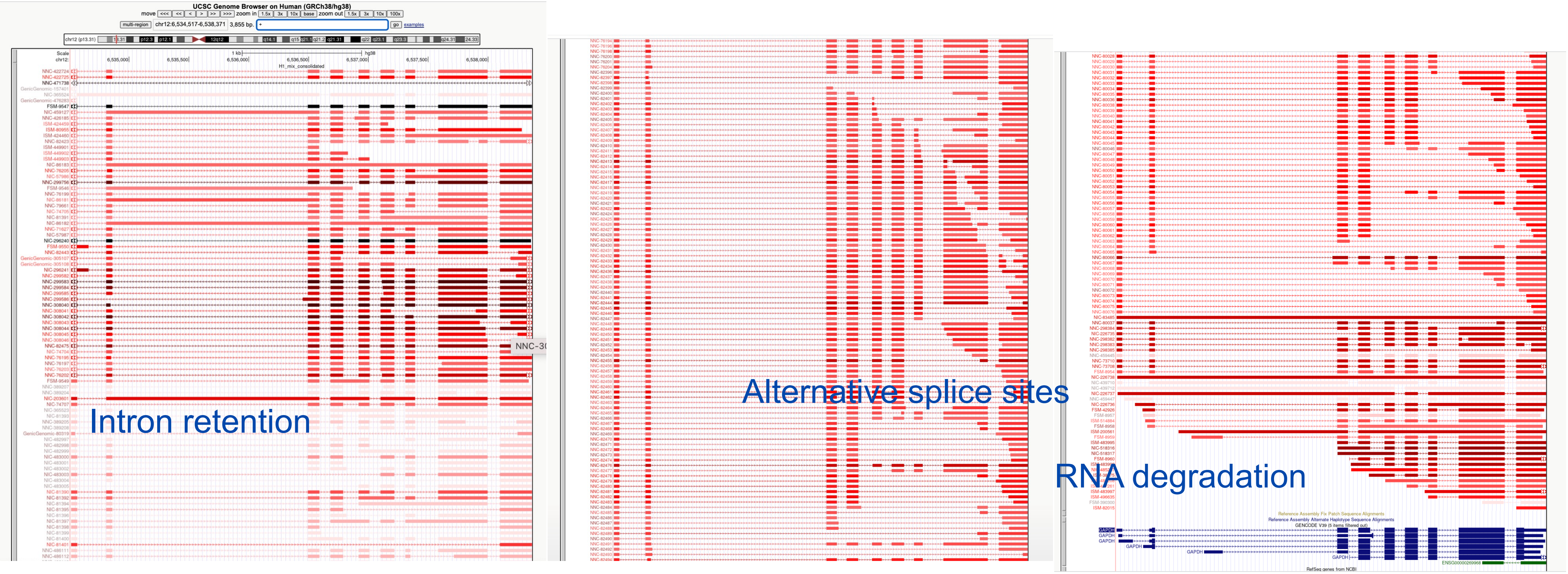
Alternative splice sites



RNA degradation

Transcripts follow different splicing patterns

A glance on the actual challenge – Transcript models for GADPH gene



Transcripts follow different splicing patterns

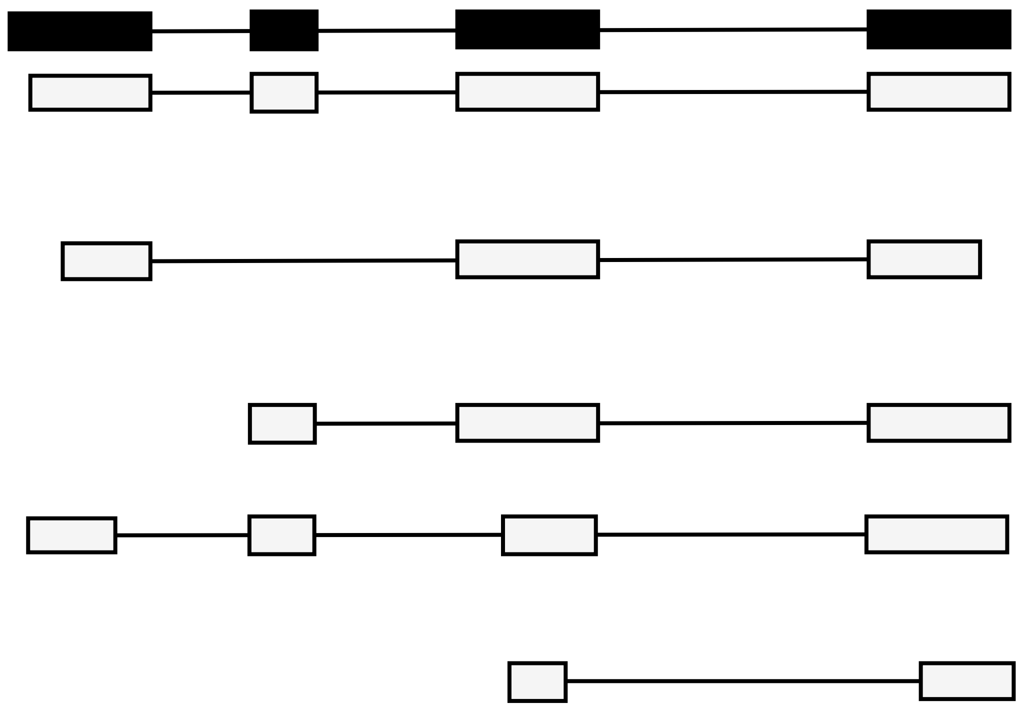
Solution: Transcript models

Reference transcript

- Millions of reads, many minor differences
- **Goal**: collapse similar reads to **identify** consistently occurring **transcript models**

Differentiate biological and technical artifacts from real transcripts

Solution: Transcript models



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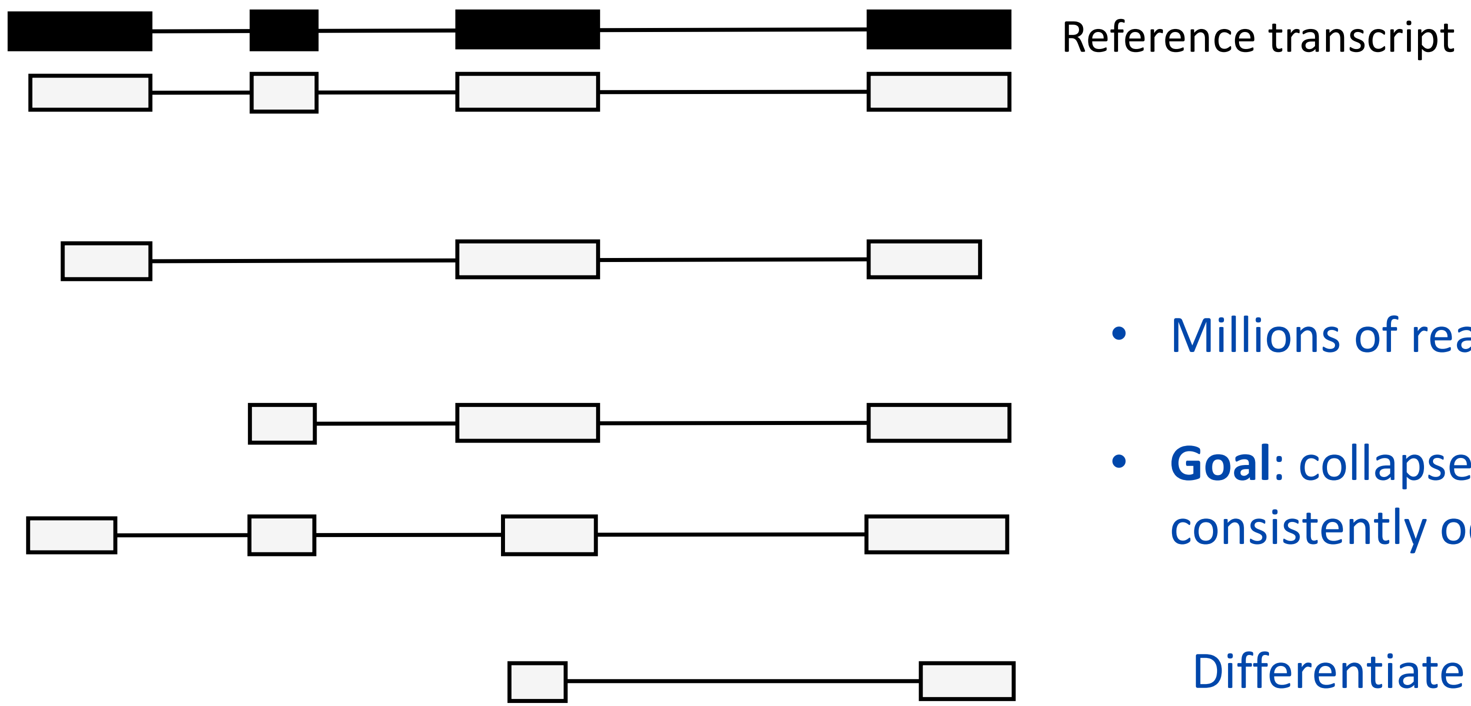


- Millions of reads, many minor differences
- **Goal**: collapse similar reads to **identify** consistently occurring **transcript models**

Differentiate biological and technical artifacts from real transcripts

This process is also called:

- Transcript | isoform identification | discovery
- Transcriptome reconstruction
- Transcript(ome) assembly (inherited from short-reads)

Getting started!

Different sequencer, different pre-processing

PacBio

CSS (raw reads)
↓
FL.bam (unmapped)
↓
FLNC.bam (unmapped)
↓
clustered.bam (unmapped)
↘

Oxford NANOPORE Technologies

Pod5 (signal)
↓
Raw.fastq
↓
Trimmed.fastq
↙

Mapping

Getting started!

Different sequencer, different pre-processing

PacBio

https://isoseq.how/

Iso-Seq

Scalable
De Novo
Isoform Discovery
from PacBio HiFi Reads

Getting Started

Recommended bulk Iso-Seq workflow

Command	Description	Output format
lima	Remove cDNA primers	`fl.bam`
isoseq refine	Remove polyA tail and artificial concatemers	`flnc.bam`
isoseq cluster2	*De novo* isoform-level clustering scalable to large number of reads (e.g. 40-100M FLNC reads)	`clustered.bam`

Oxford NANOPORE Technologies

EPI2ME Documentation

Documentation
- wf-alignment
- wf-amplicon
- wf-artic
- wf-bacterial-genomes
- wf-basecalling
- wf-cas9
- wf-clone-validation
- wf-flu
- wf-human-variation
- wf-metagenomics
- wf-mpx
- wf-pore-c
- wf-single-cell
- wf-somatic-variation
- wf-tb-amr
- wf-teloseq
- wf-template

Transcriptomes

Transcriptome analysis of cDNA and direct RNA sequencing data.

Introduction

This workflow can be used for the following:

- Identify RNA transcripts using either cDNA or direct RNA reads.
- Reference aided transcriptome assembly.
- Annotation of assembled transcripts.
- Differential gene expression analysis using a pre-computed or assembled reference transcriptome.
- Differential transcript usage analysis using a precomputed or assembled reference transcriptome.

https://epi2me.nanoporetech.com/

nextflow

epi2me-labs / **wf-transcriptomes**

Code | Issues 14 | Pull requests | Agents

wf-transcriptomes Public

Raw PacBio Data

Deepconsensus

PacBio

CSS (raw reads)

↓

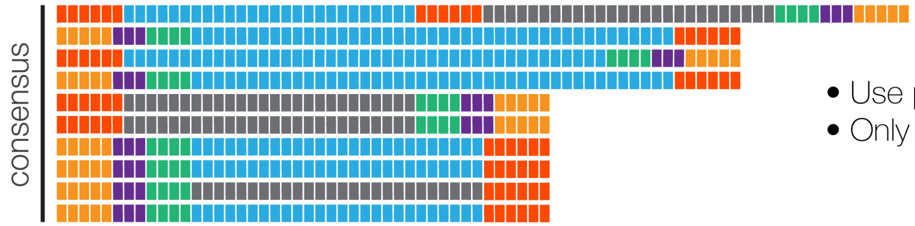
FL.bam (unmapped)

↓

FLNC.bam (unmapped)

↓

clustered.bam (unmapped)



- Use **polished** CCS reads
- Only full-pass ZMWs

https://isoseq.how/getting-started.html

Demultiplexed full-length reads

lima

PacBio

CSS (raw reads)

↓ Deepconsensus + lima

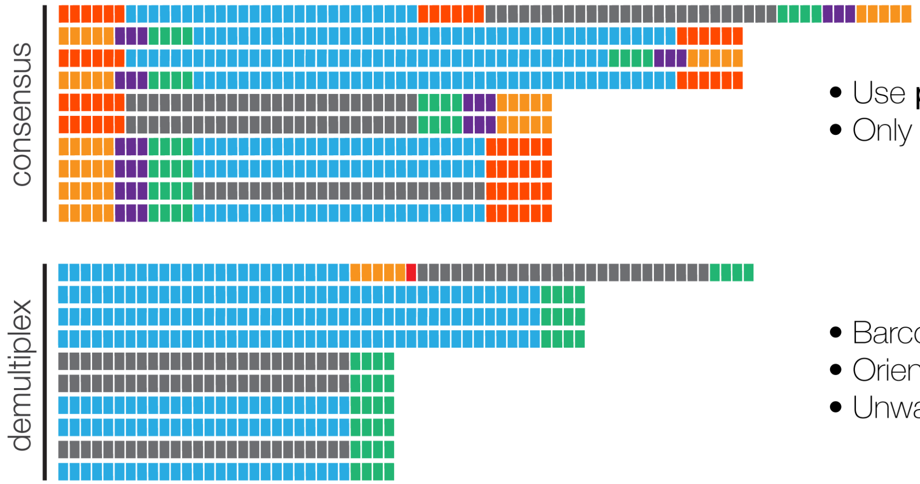
FL.bam (unmapped)

↓

FLNC.bam (unmapped)

↓

clustered.bam (unmapped)



- Use **polished** CCS reads
- Only full-pass ZMWs

- Barcoded and unbarcoded cDNA primer removal
- Orientation
- Unwanted primer combination removal

https://isoseq.how/getting-started.html

Processing PacBio sequencing

Isoseq refine

CSS (raw reads)

Deepconsensus + lima ↓

FL.bam (unmapped)

isoseq refine ↓

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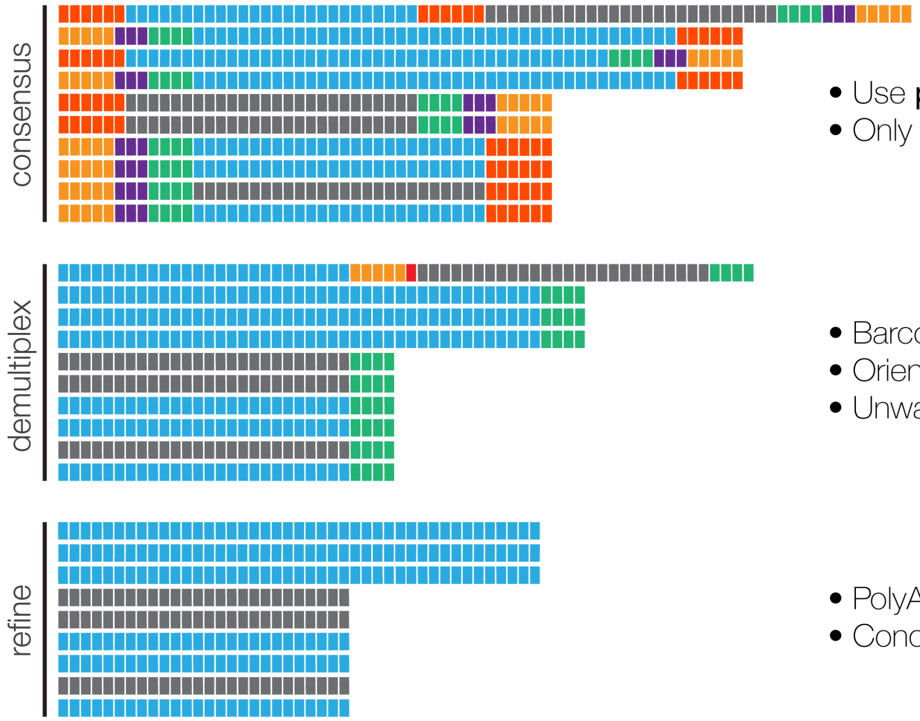
↓

clustered.bam (unmapped)



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- Barcoded and unbarcoded cDNA primer removal
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- Unwanted primer combination removal

- PolyA tail trimming
- Concatemer removal

https://isoseq.how/getting-started.html

PacBio ready to map!

Isoseq cluster2

PacBio

CSS (raw reads)

↓ Deepconsensus + lima

FL.bam (unmapped)

↓ isoseq refine

FLNC.bam (unmapped)

↓ isoseq cluster2

clustered.bam (unmapped)

- Use **polished** CCS reads
- Only full-pass ZMWs

- Barcoded and unbarcoded cDNA primer removal
- Orientation
- Unwanted primer combination removal

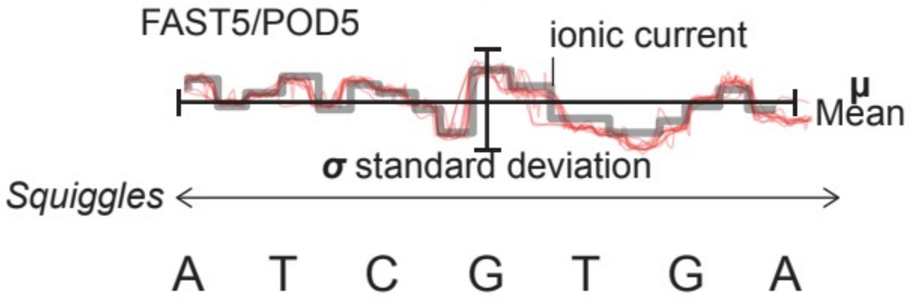
- PolyA tail trimming
- Concatemer removal

- Hierarchical, n*log(n) clustering, alignment of shorter to longer sequences
- Iterative cluster merging
- Generate consensus for each read cluster using QV guided PoA

https://isoseq.how/getting-started.html

Raw Nanopore data

Basecall



Dorado

Pod5 (signal)

↓ Dorado

Raw.fastq

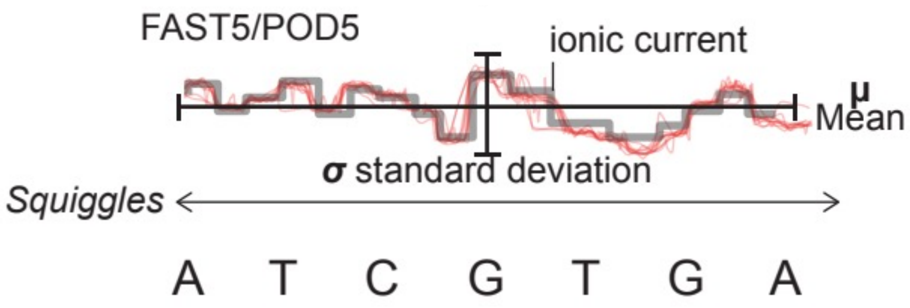
↓

Trimmed.fastq

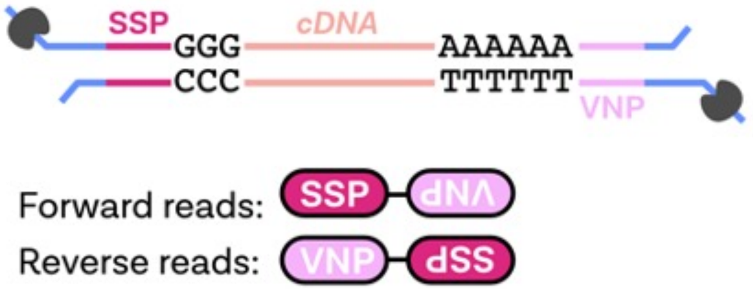
https://epi2me.nanoporetech.com/wfindex/

Processing Nanopore data

Basecall



Identify, orient, trim polyA, rescue chimeric reads



Pychopper

Pod5 (signal)

↓ Dorado

Raw.fastq

↓ Pychopper

Trimmed.fastq

https://epi2me.nanoporetech.com/wfindex/

Ready to map!

Different sequencer, different pre-processing

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Dorado

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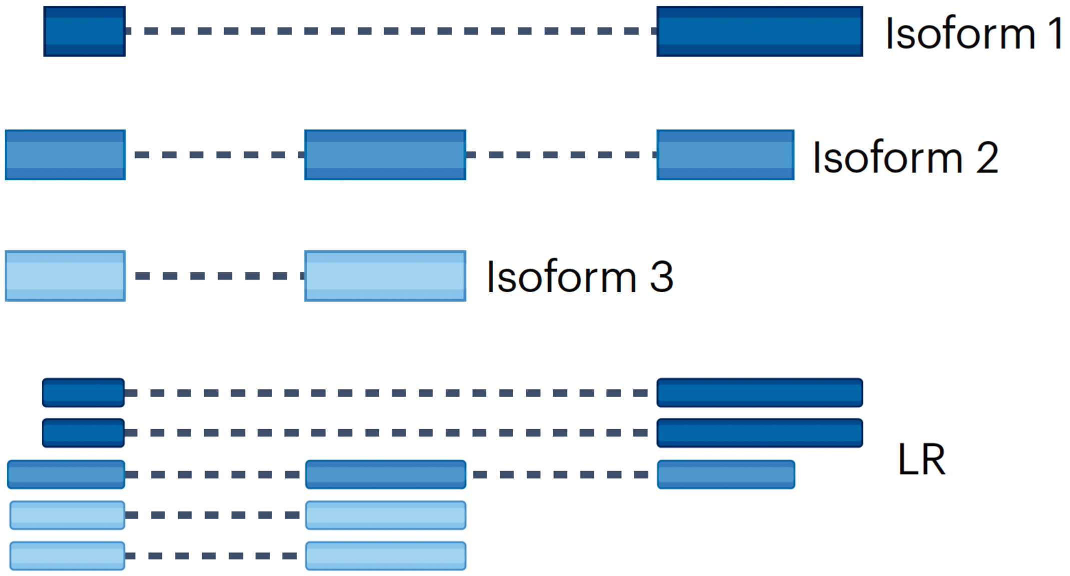
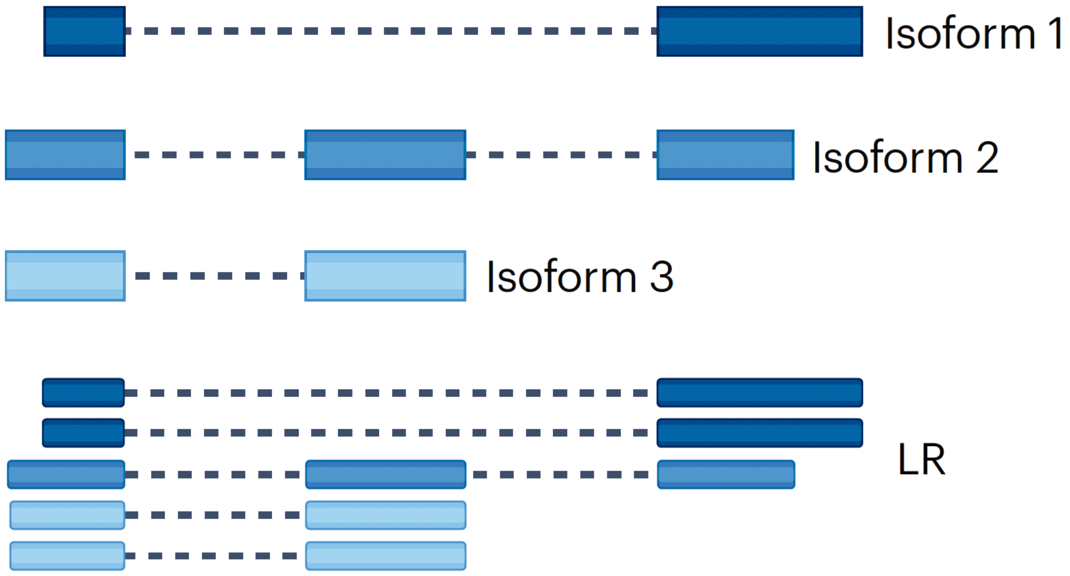
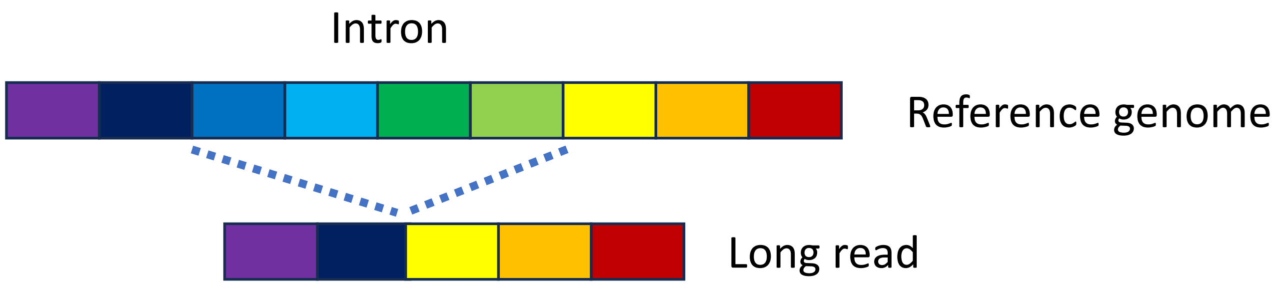
Mapping

Mapping: minimap2



Mapping: minimap2

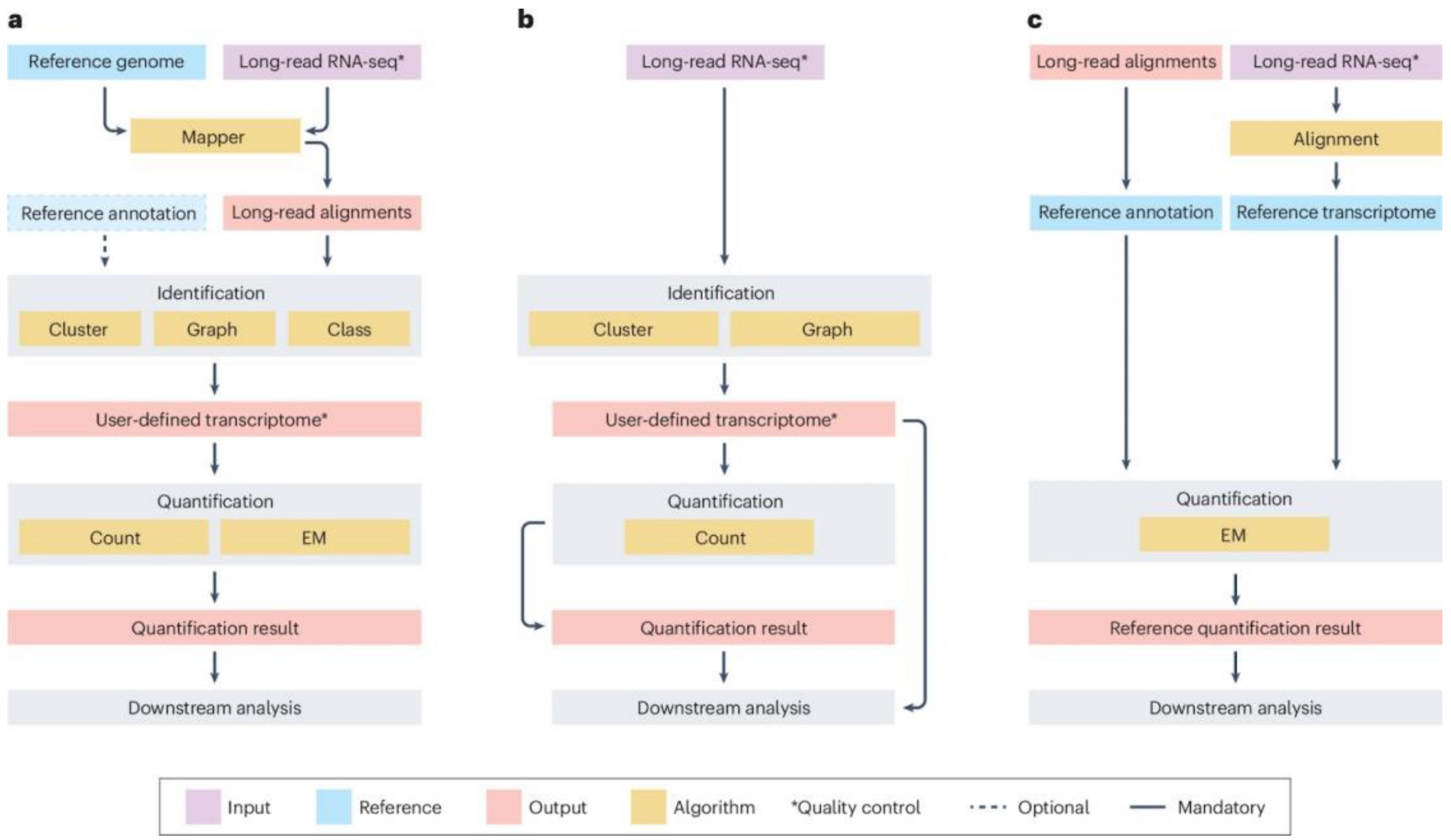




Spliced mapper: minimap2

Isoform identification and quantification strategies

Reference guided **Reference-free (de novo)** **Direct quantification**



Reference guided

- Identify isoforms based on
  - Reference genome
  - Reference annotation
- IsoTools, IsoQuant, FLAIR, Bambu etc
- Restrictive or permissive

Reference free

- Without any reference information
- isON-pipeline, RNA-bloom2, RATTLE etc

Direct quantification

- No discovery of novel isoforms
- Oarfish, lr-kallisto, LIQA, Nanocount etc

Figure source: Monzó, Liu, Conesa. 2025. 10.1038/s41576-025-00828-z

Transcript reconstruction: graph

IsoQuant

Alignment to reference genome

Transcriptome reconstruction

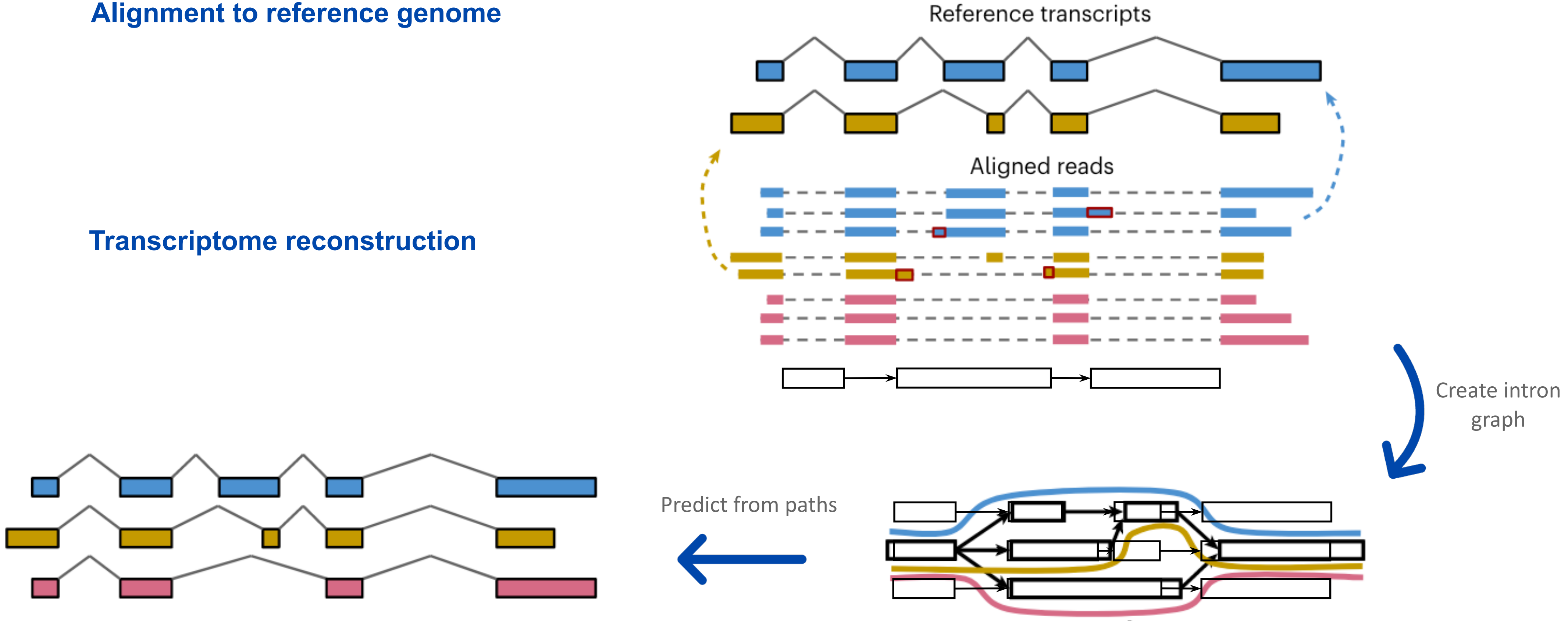


Figure modified from IsoQuant paper.

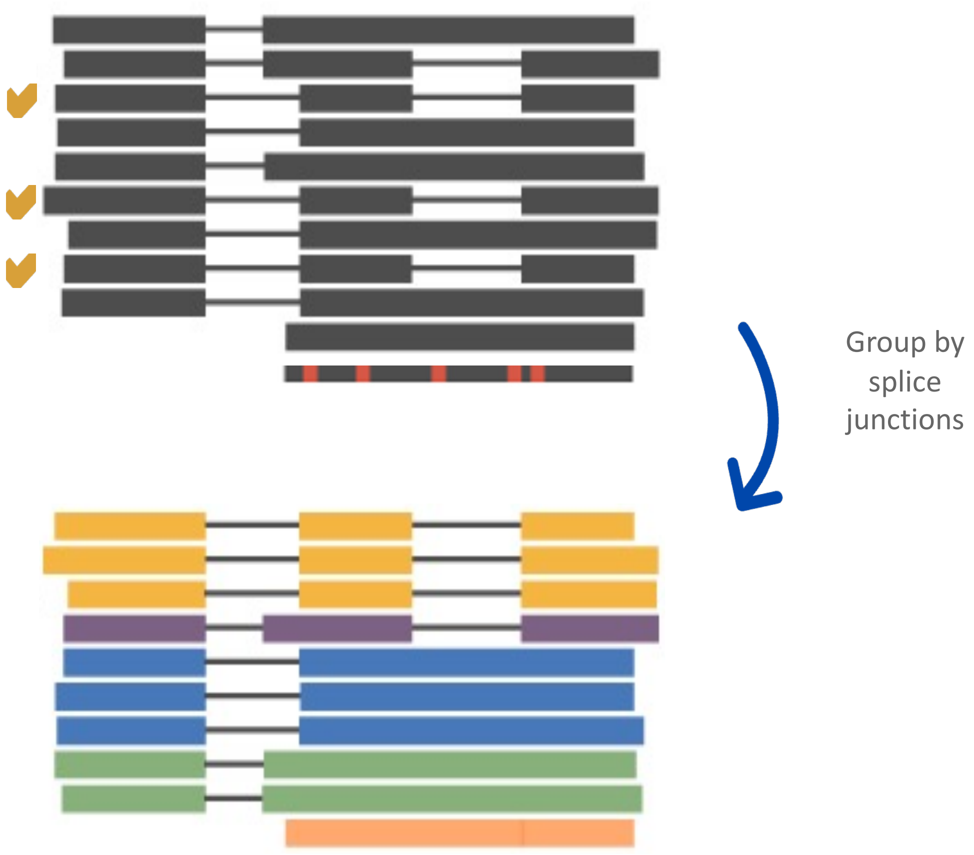
Prjibelski, Andrey D et al. Nature biotechnology vol. 41,7 (2023): 915-918. doi:10.1038/s41587-022-01565-y

Transcript reconstruction: cluster

Flair

Alignment to reference genome

Transcriptome reconstruction



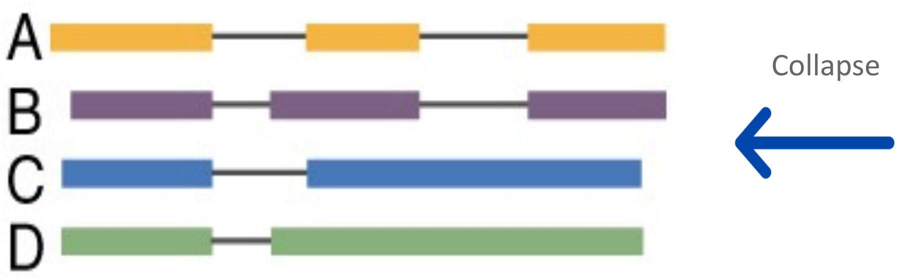


Figure modified from Flair paper.

Flair: Tang, Alison D et al. Nature communications vol. 11,1 1438. 18 Mar. 2020, doi:10.1038/s41467-020-15171-6

Flair2: Tang, Alison D et al. Genome biology vol. 25,1 173. 2 Jul. 2024, doi:10.1186/s13059-024-03301-y

Reflection and recommendations

You have lrRNA-seq data of an organism that has not been sequenced before

You have lrRNA-seq data of a model organism but are not interested in discovering novel isoforms

You have lrRNA-seq data of a model organism and are interested in condition-specific differences in alternative splicing, including novel isoforms

You have a good reference transcriptome and only care about quantification

Reflection and recommendations

You have lrRNA-seq data of an organism that has not been sequenced before

Reference-free strategies, high quality data, replicates, orthogonal data (isON-pipeline, RNA-Bloom)

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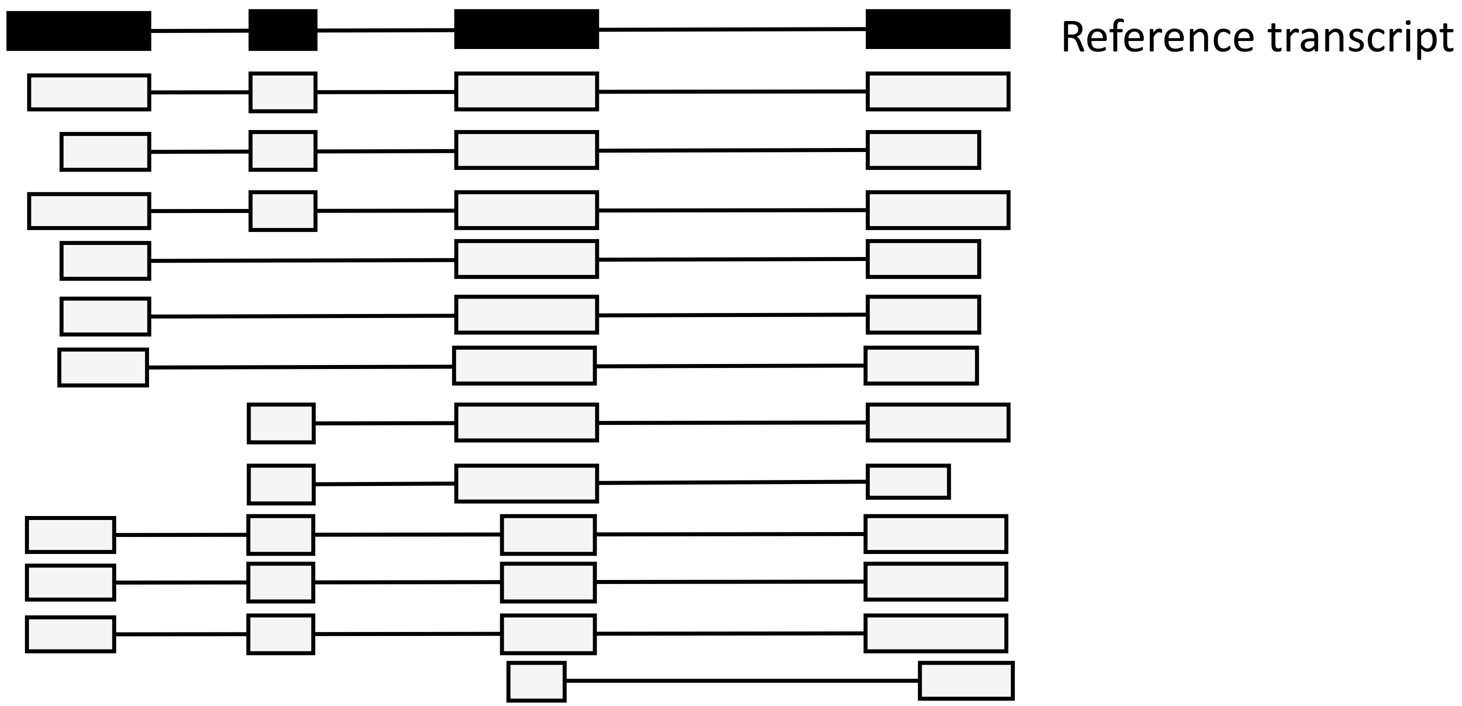
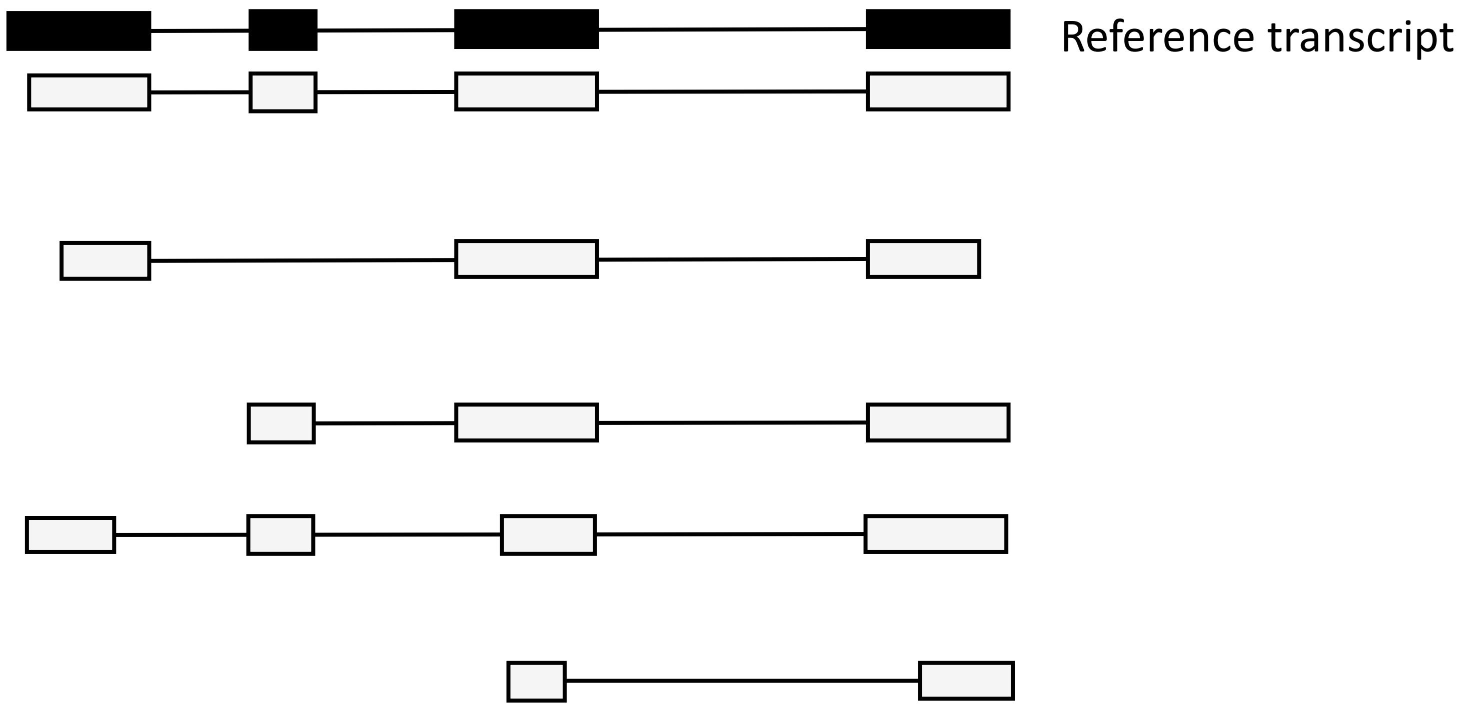
You have a good reference transcriptome and only care about quantification

Direct quantification (lrKallisto, Oarfish)

From raw reads

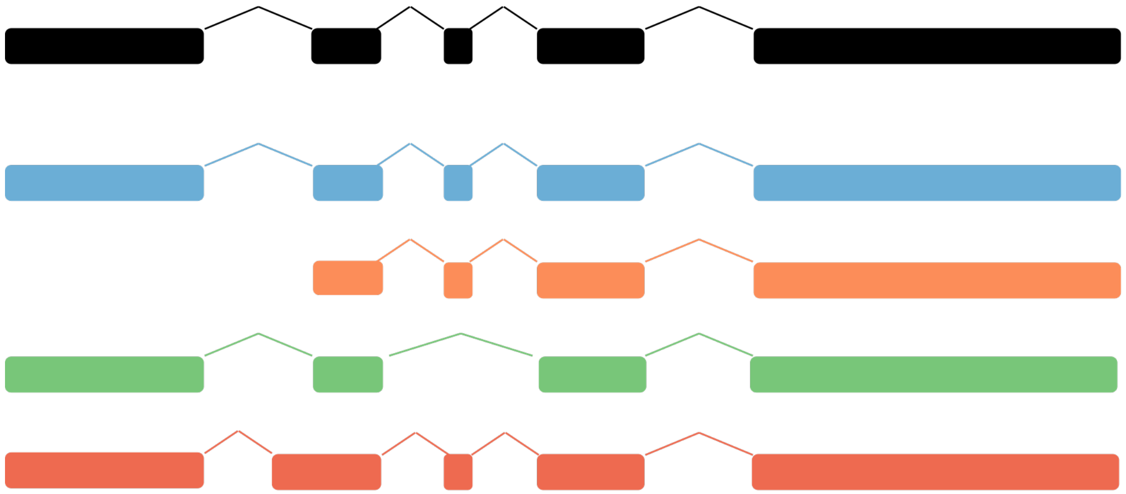


To transcript models



Quality Control!

SQANTI3



REFERENCE TRANSCRIPT

Full **S**plice **M**atch (**FSM**)
Matches all SJ perfectly

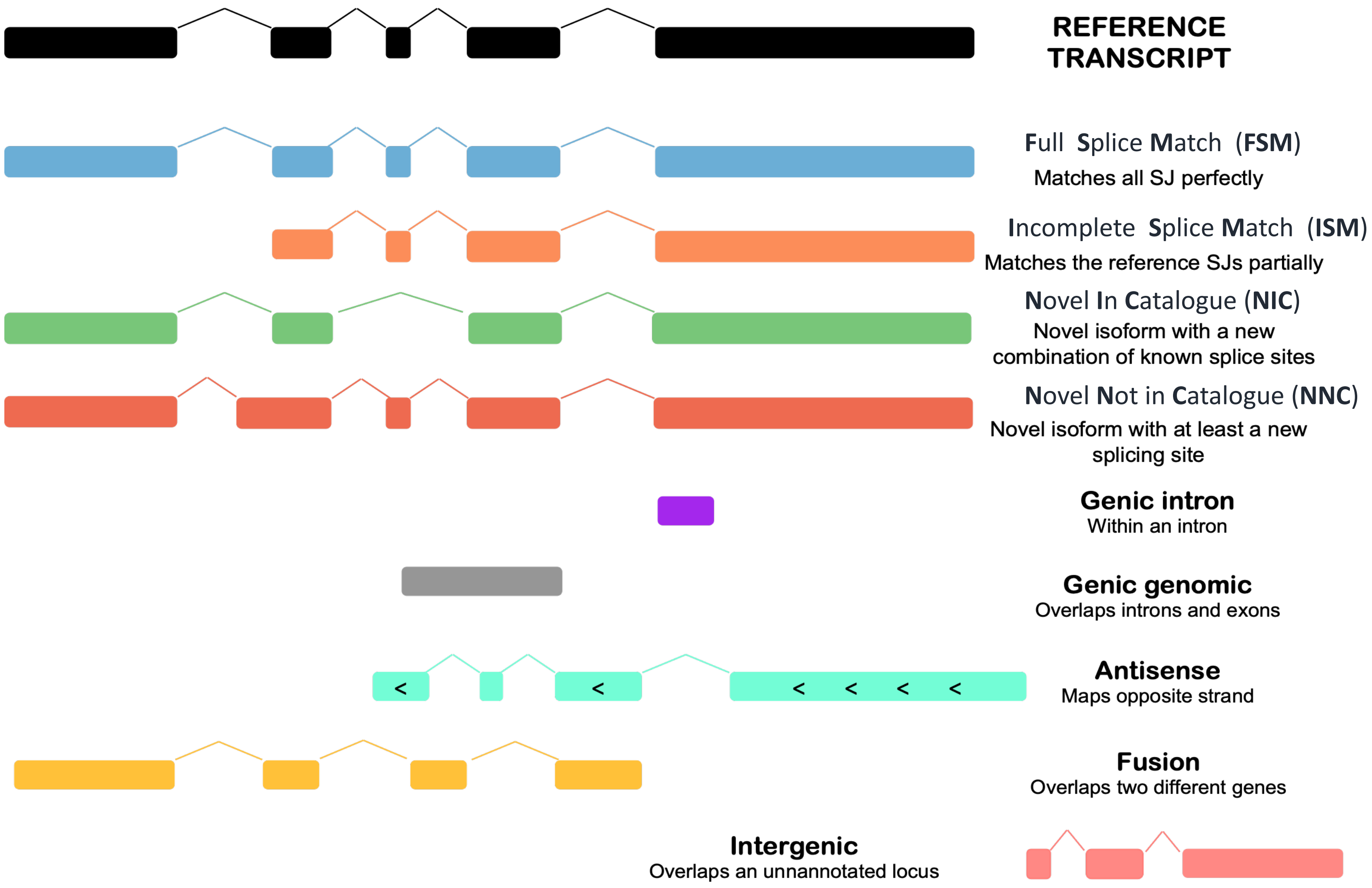
Incomplete **S**plice **M**atch (**ISM**)
Matches the reference SJs partially

Novel **I**n **C**atalogue (**NIC**)
Novel isoform with a new combination of known splice sites

Novel **N**ot in **C**atalogue (**NNC**)
Novel isoform with at least a new splicing site

Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

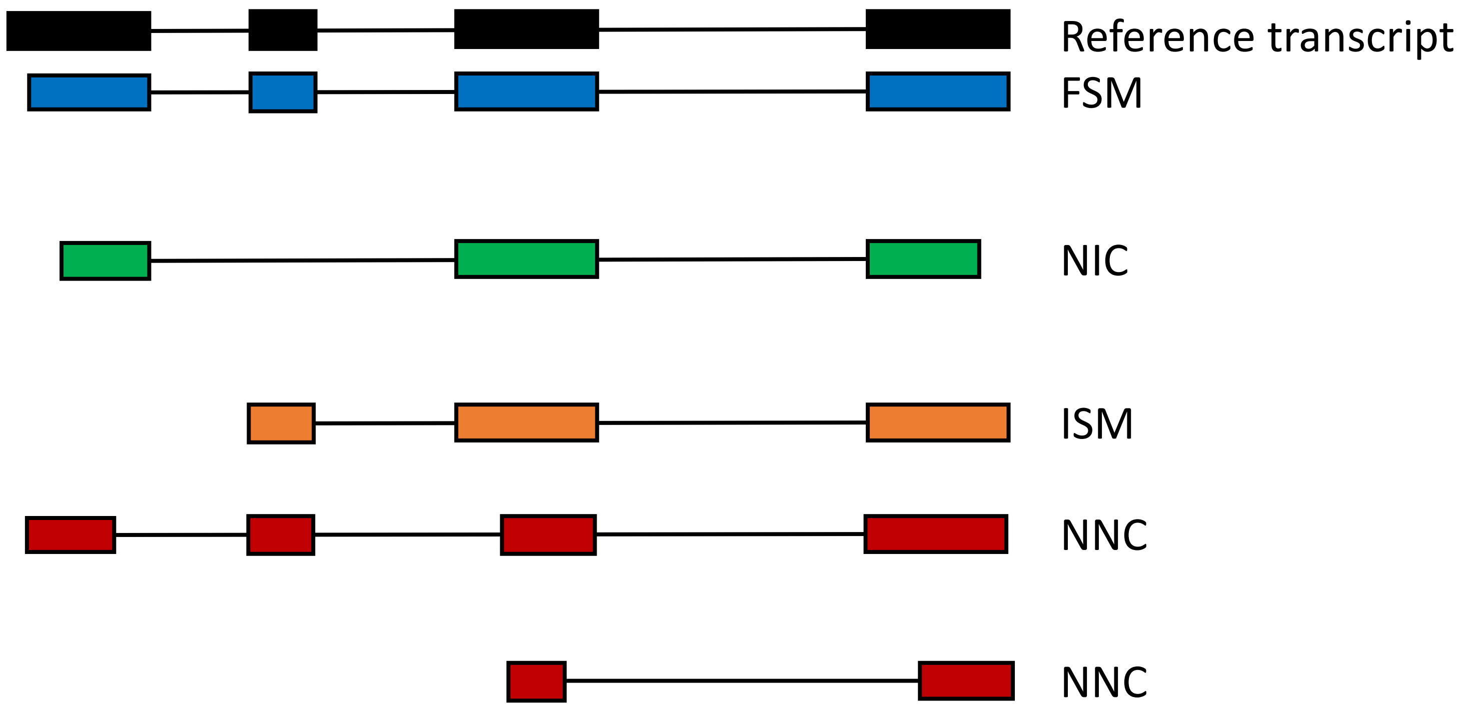
Quality Control!



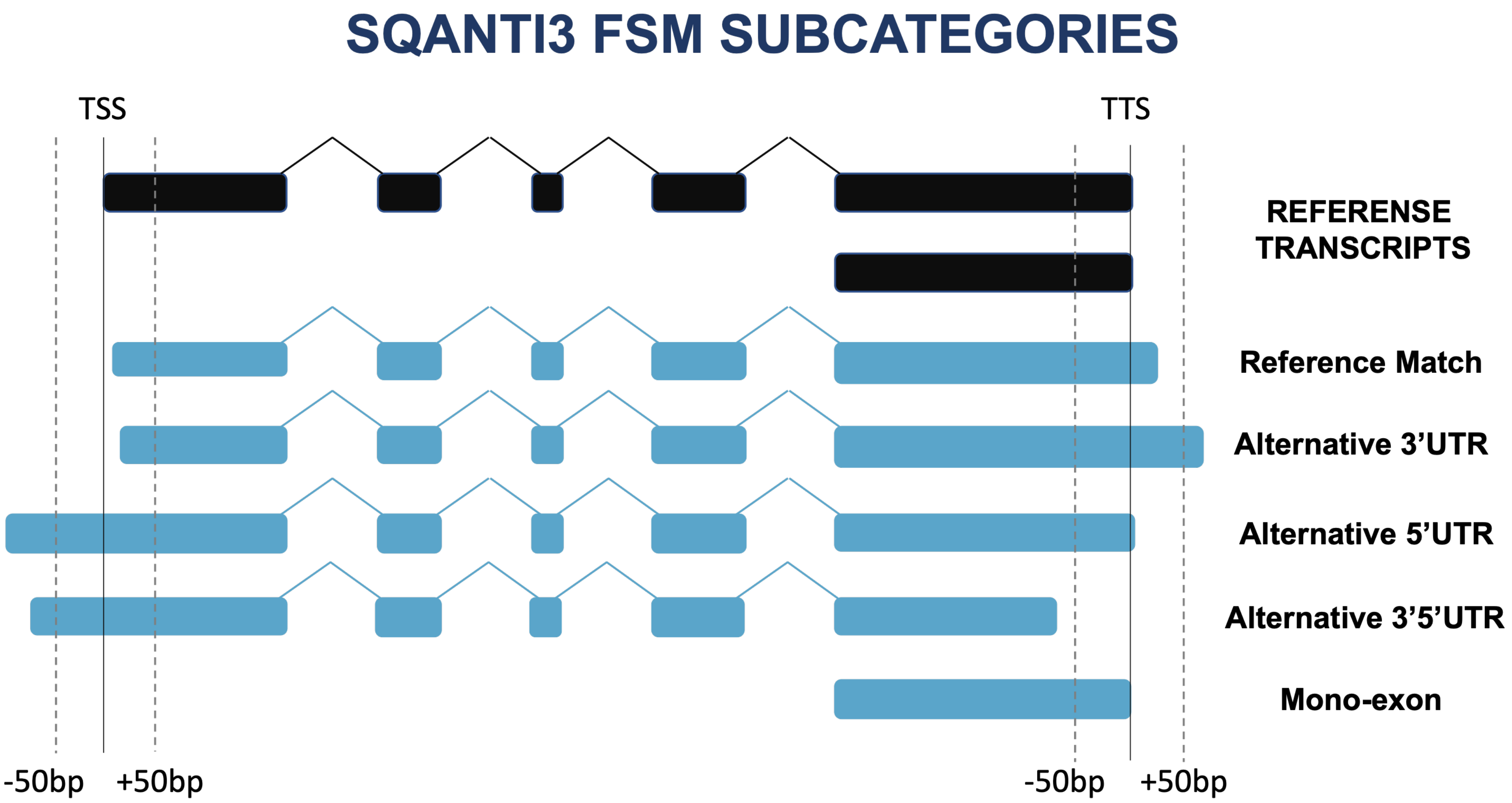
Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

To transcript models

SQANTI3



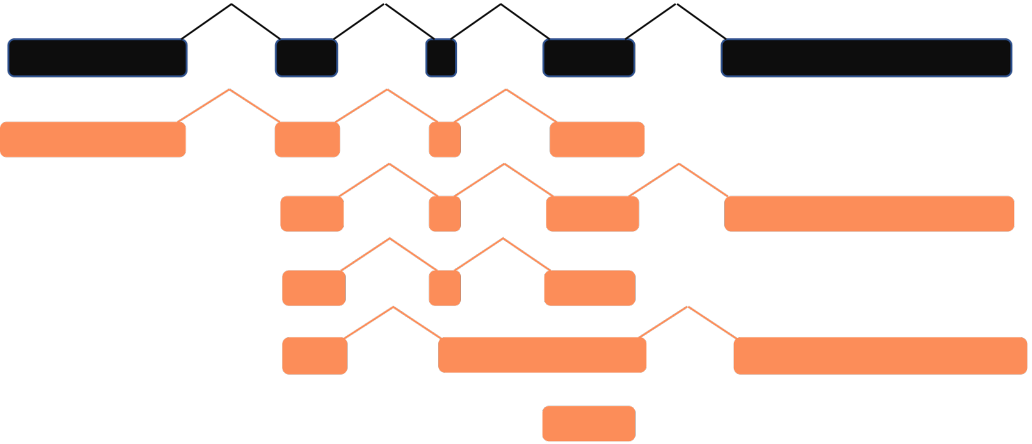
Novel FSM categories in SQANTI3



Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

Other novel categories in SQANTI3

SQANTI3 ISM SUBCATEGORIES



REFERENSE TRANSCRIPT

ISM 5' fragment

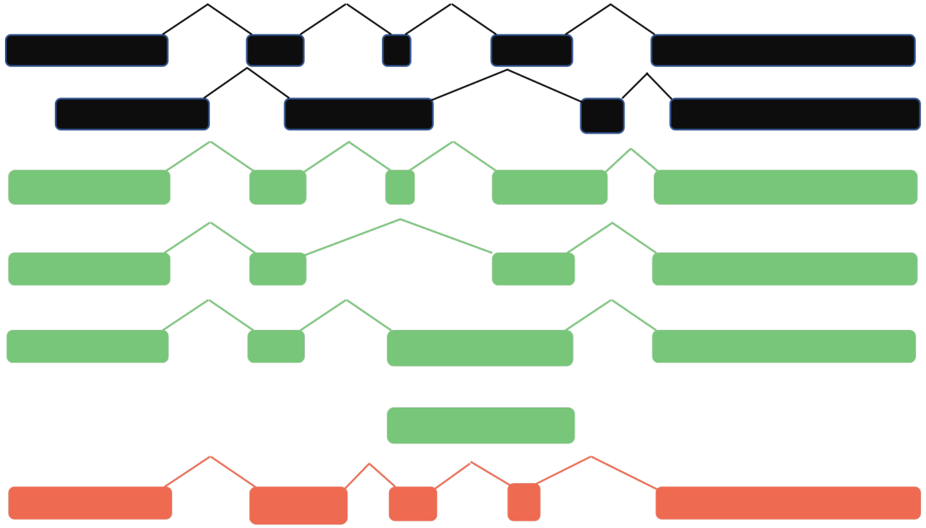
ISM 3' fragment

ISM internal fragment

ISM intron retention

ISM mono-exon

SQANTI3 NIC AND NNC SUBCATEGORIES



REFERENSE TRANSCRIPTS

NIC
Combination of known SJ

NIC
Combination of known splice sites

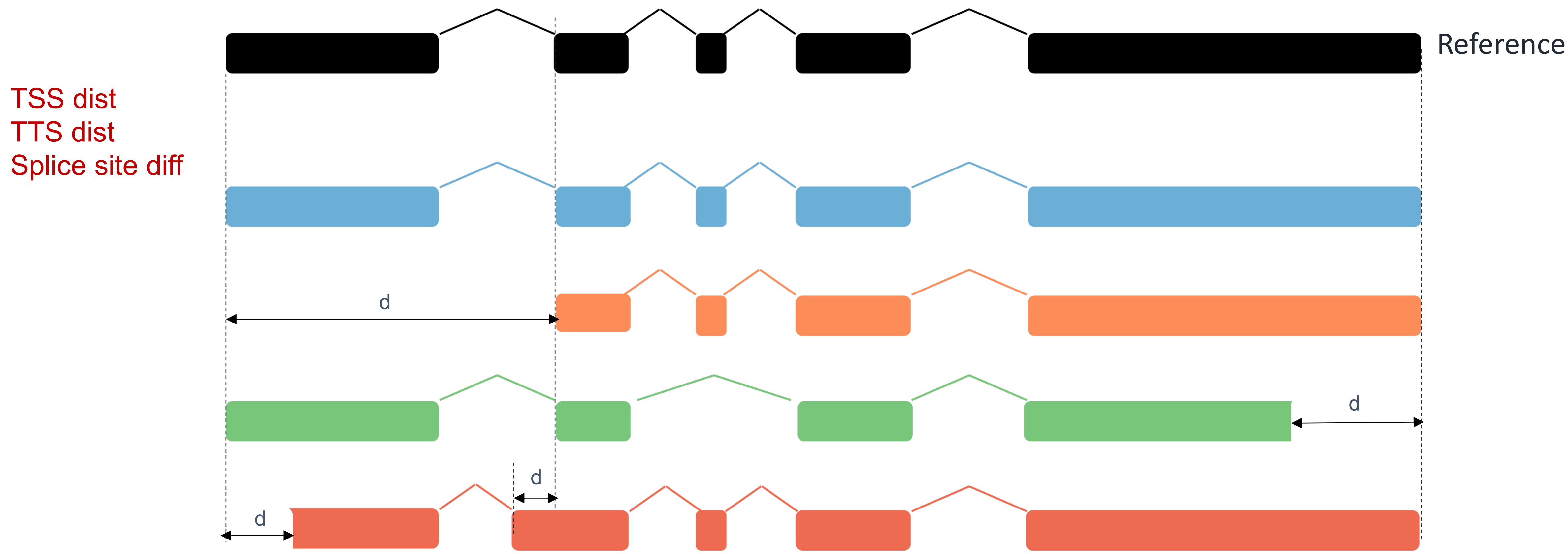
NIC
Intron retention

NIC
Mono-exon by intron retention

NNC
At least one novel splice site

Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

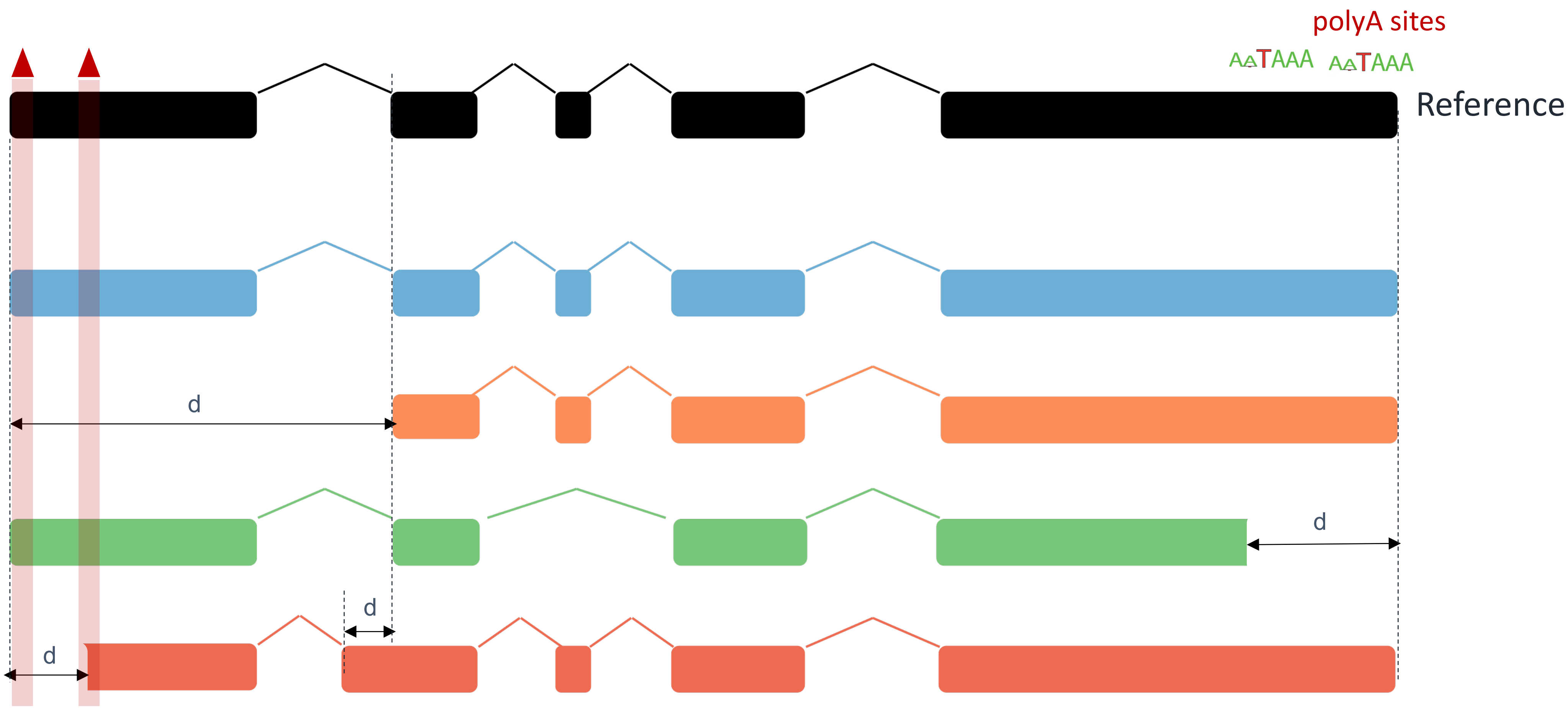
Distance to TSS, TTS and to known junctions



Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

CAGE and polyA support

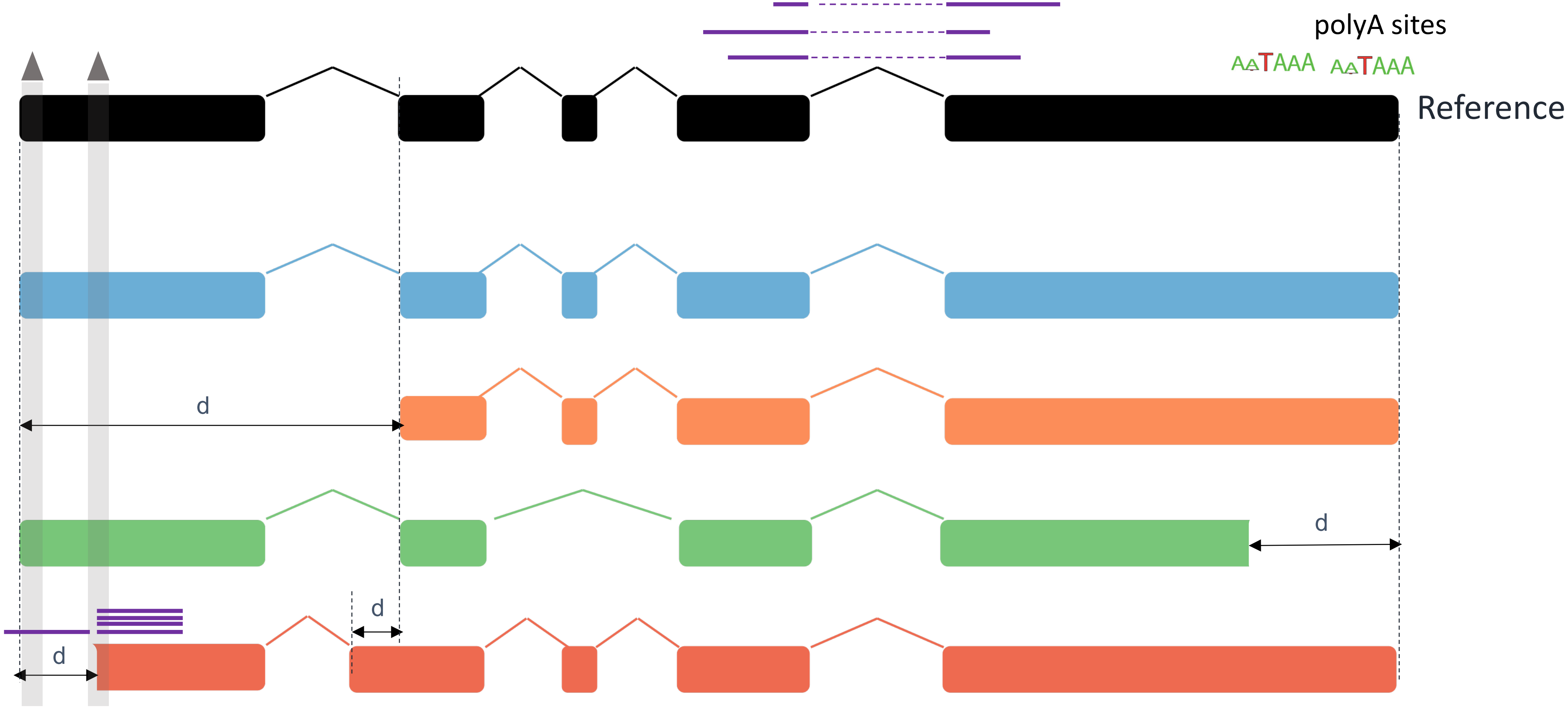
TSS dist
TTS dist
Splice site diff
CAGE support
PolyA support



Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

Short reads and lrRNA-seq support

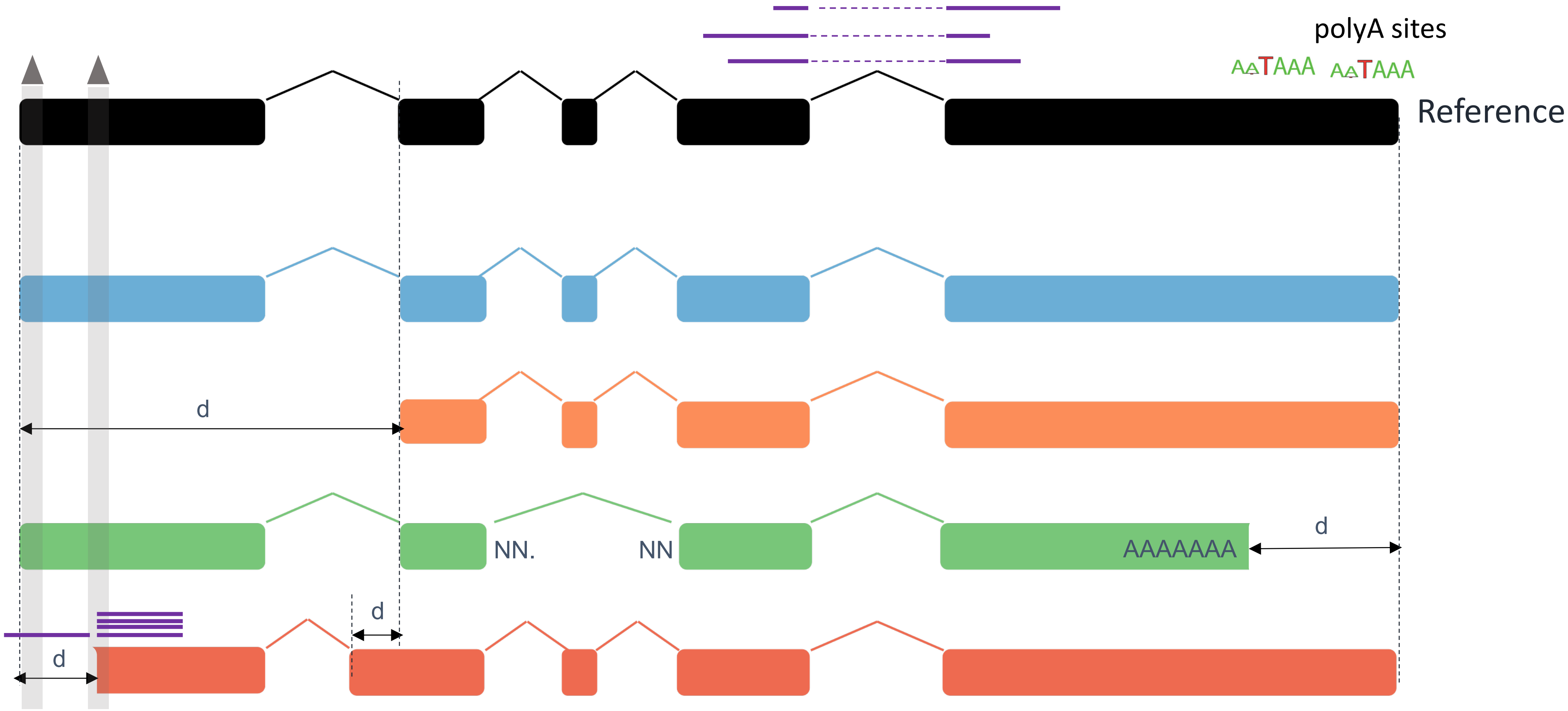
TSS dist
TTS dist
Splice site diff
CAGE support
PolyA support
SR junctions
TSS ratio
SR expression
FL expression
Fractional expr.



Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

Canonical junction, intraprimming, RT-switching etc

TSS dist
TTS dist
Splice site diff
CAGE support
PolyA support
SR junctions
TSS ratio
SR expression
FL expression
Fractional expr.
Canonical junct.
Donor type
Acceptor type
Intrapriming
RT-switching
etc



Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

SQANTI3

 # Genes: 18865
 # Isoforms: 52292

Gene classification

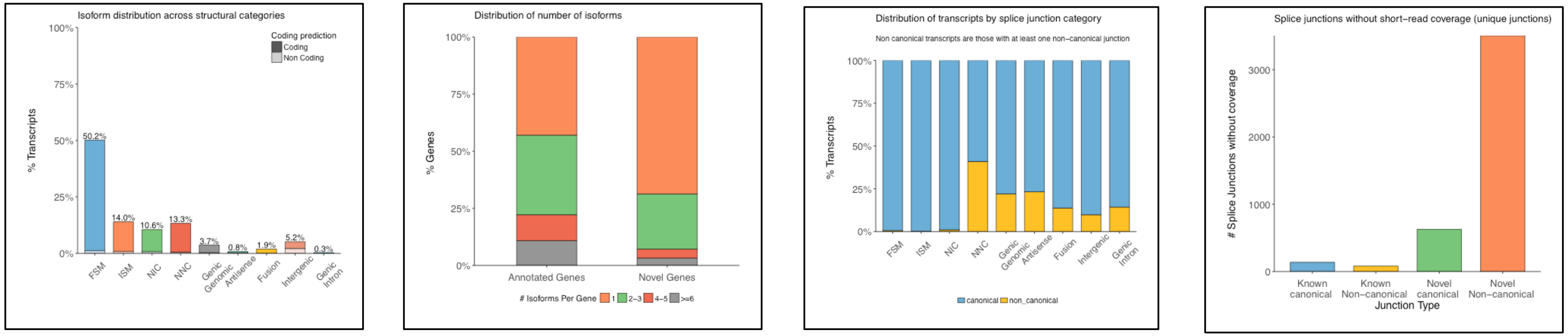
category	# genes
Annotated Genes	16991
Novel Genes	1874

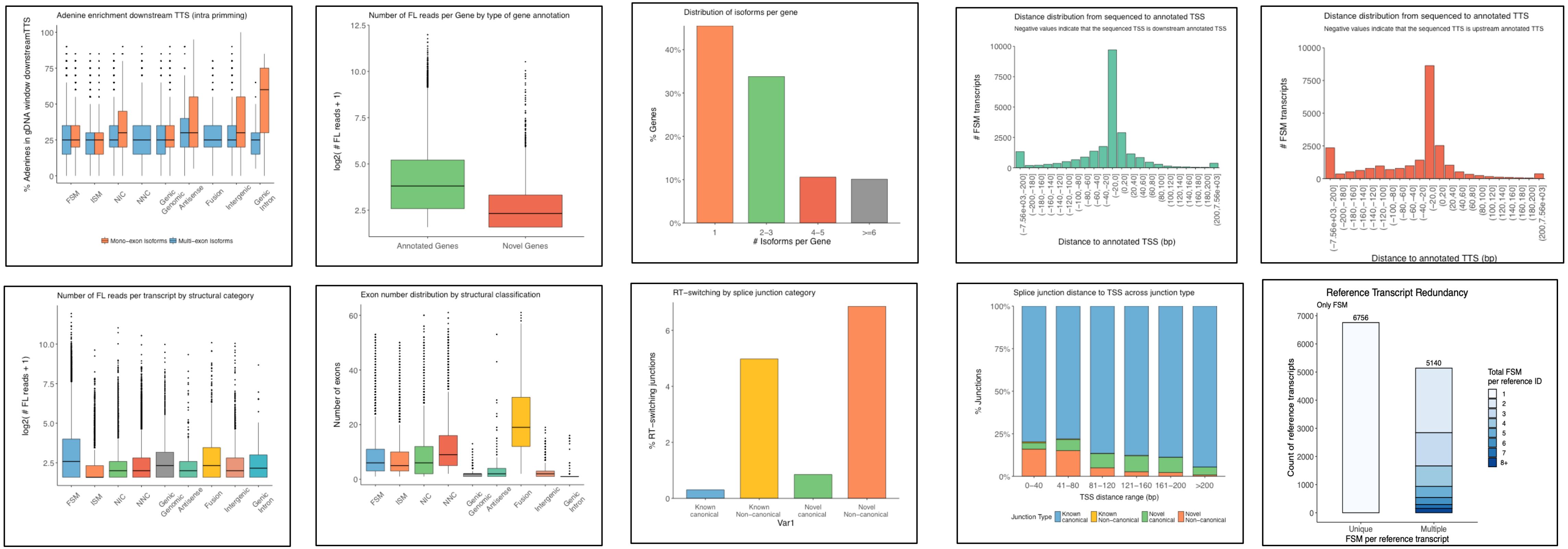
SJ classification

category	# SJ
Known canonical	100234
Known Non-canonical	156
Novel canonical	8392
Novel Non-canonical	4026

Characterization of transcripts based on splice junctions

category	# isoforms
FSM	26238
ISM	7328
NNC	6974
NIC	5525
Intergenic	2696
Genic Genomic	1945
Fusion	1004
Antisense	422
Genic Intron	160

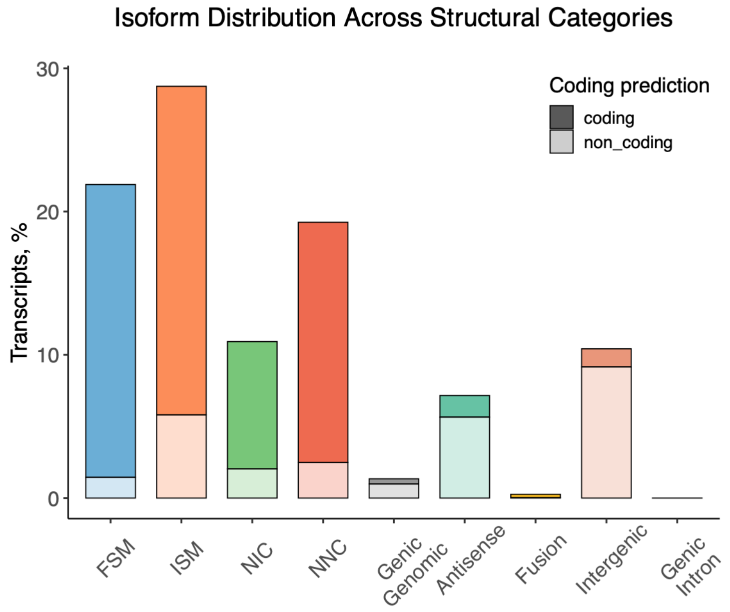




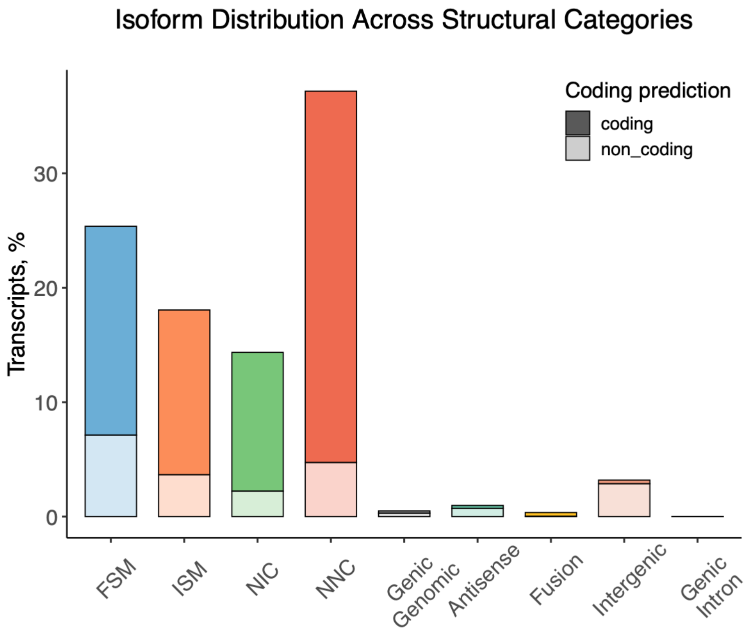
Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

Important conclusions from SQANTI3

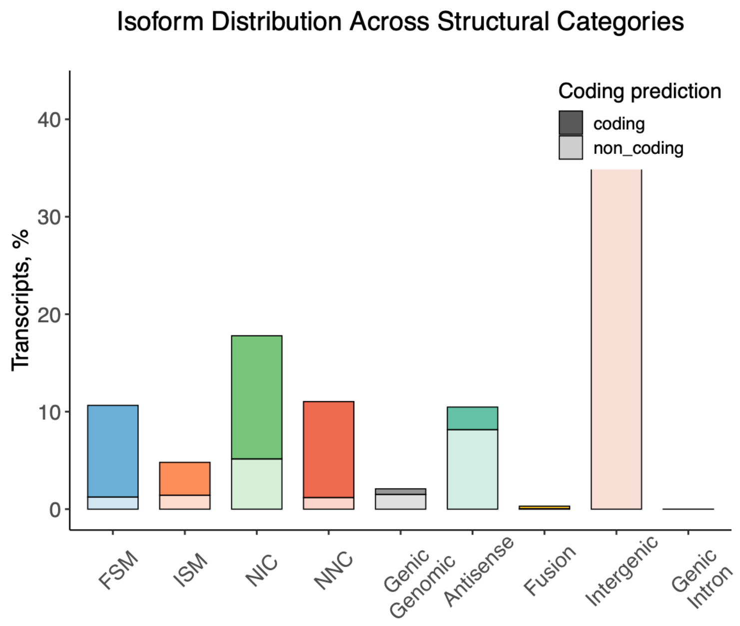
Pipeline 1



Pipeline 2

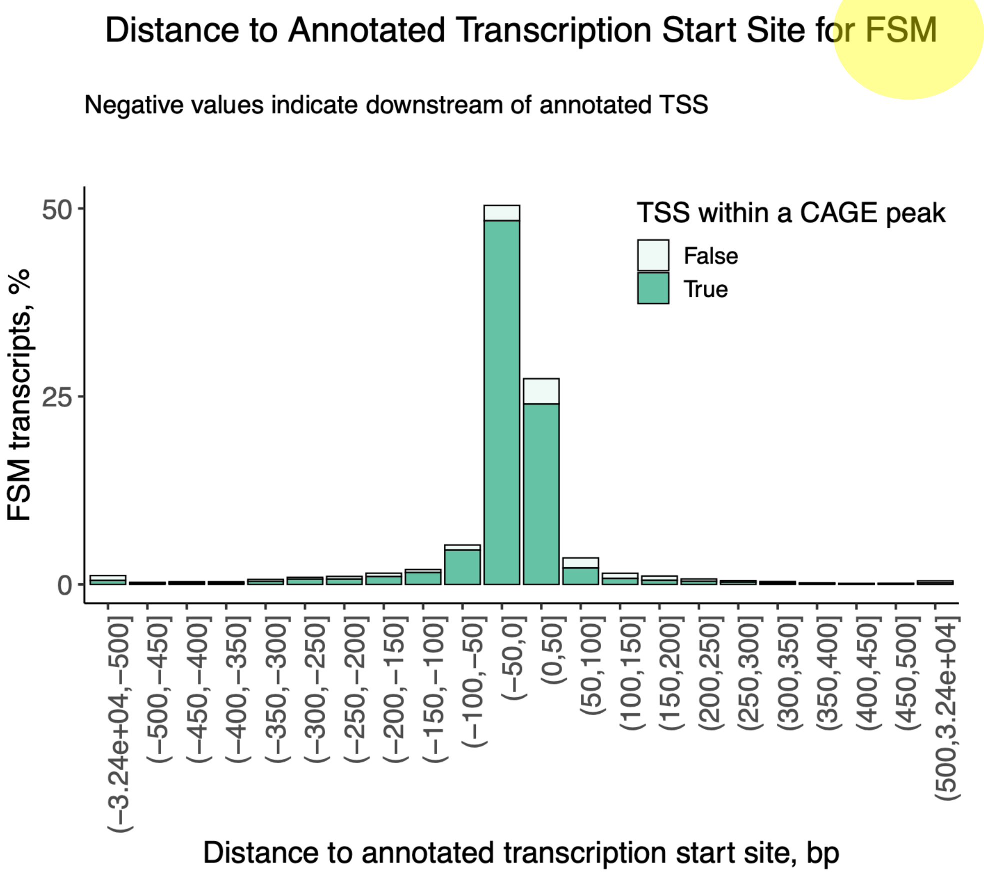


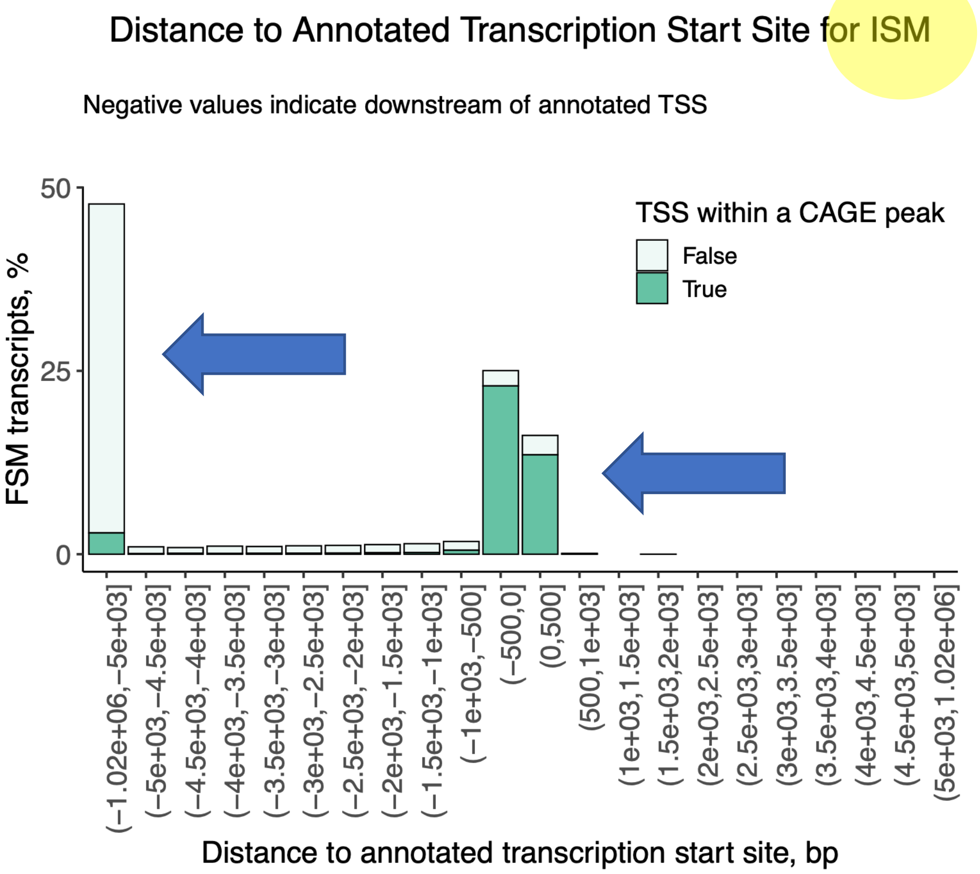
Pipeline 3



Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

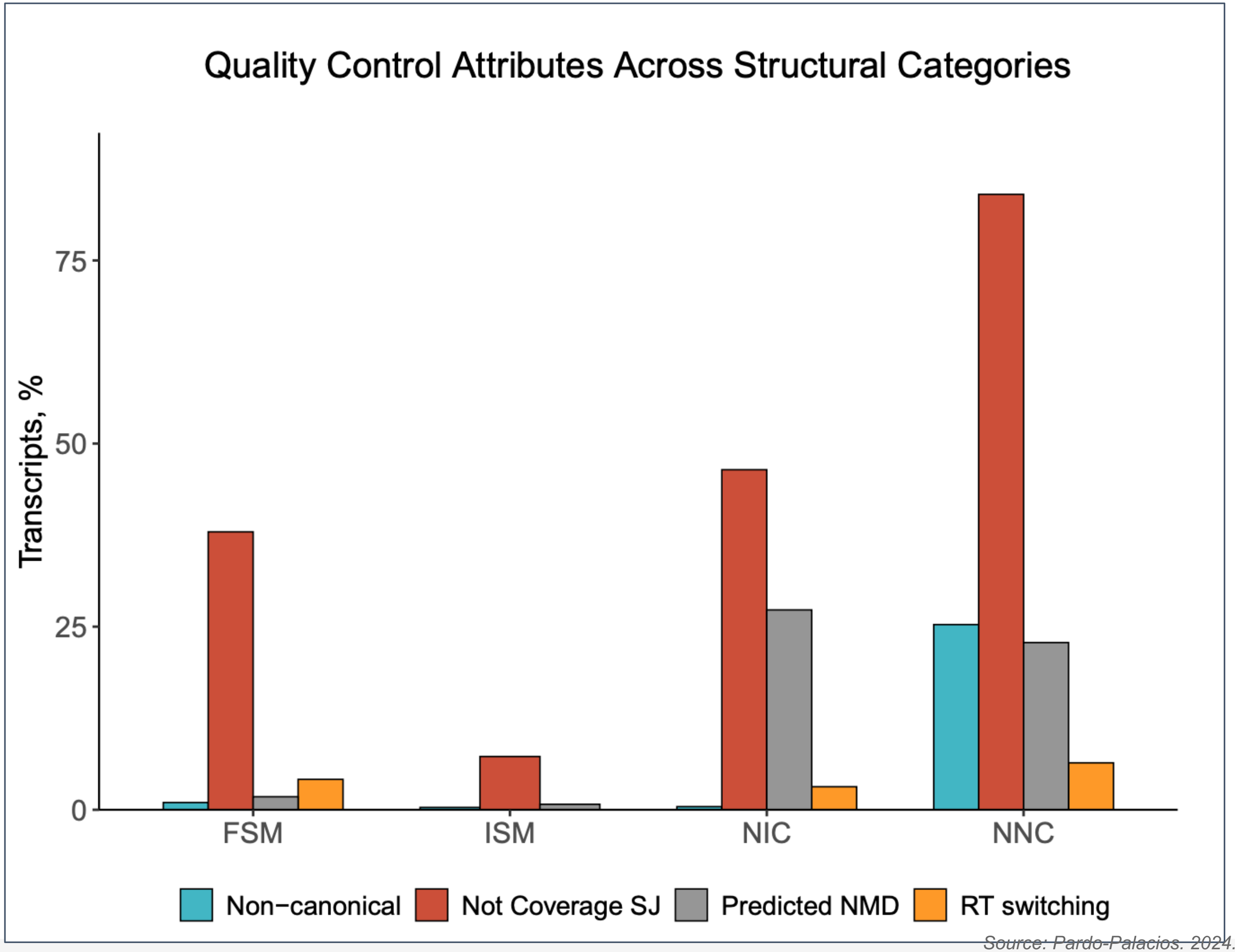
CAGE data may support FSM





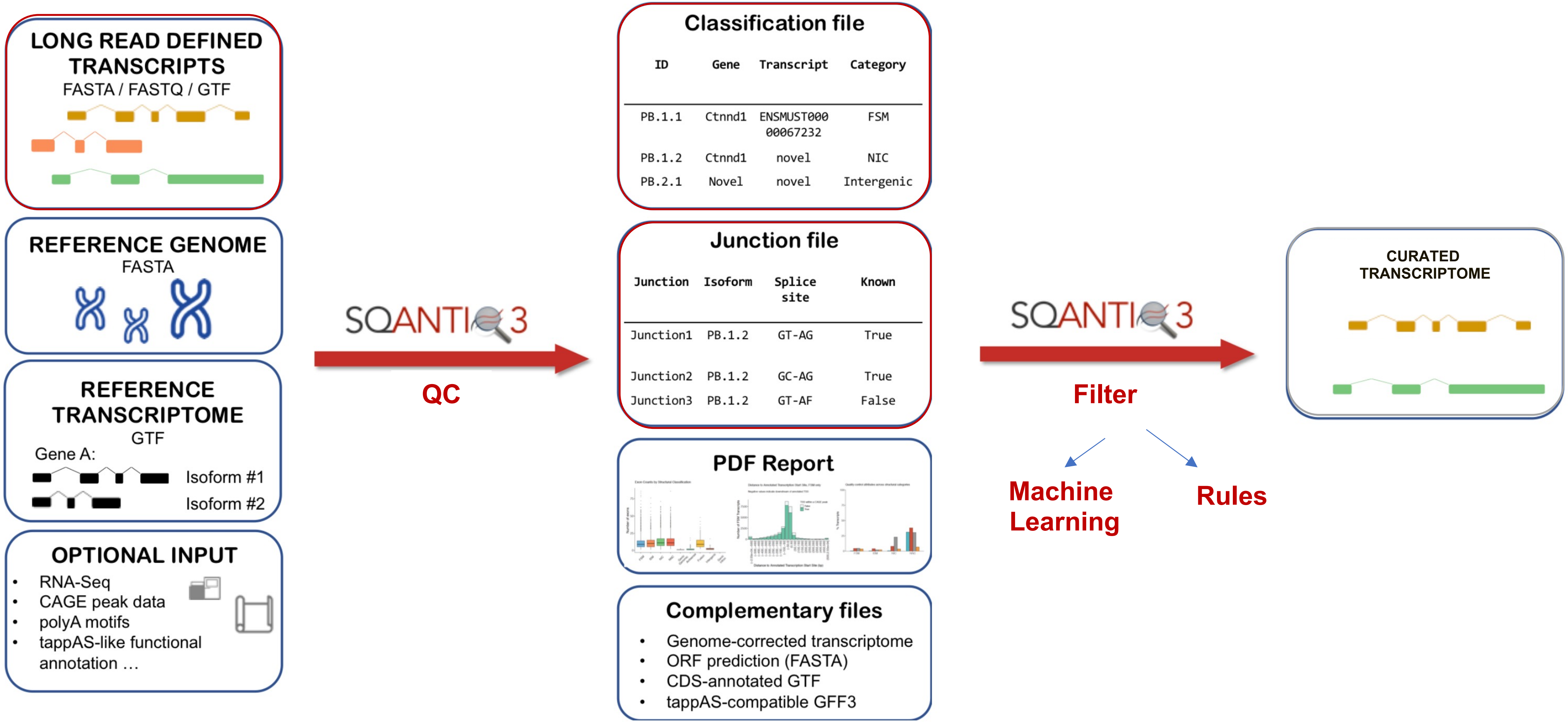
Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

Bad quality features



Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

Solving problems: SQANTI-filter



Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

Solving problems: SQANTI-filter

Rules

This filter is supplied in the `utilities/filter/filter_default.json` file:

```
{
    "full-splice_match": [
        {
            "perc_A_downstream_TTS":[0,59]
        }
    ],
    "rest": [
        {
            "perc_A_downstream_TTS":[0,59],
            "RTS_stage":"FALSE",
            "all_canonical":"canonical"
        },
        {
            "perc_A_downstream_TTS":[0,59],
            "RTS_stage":"FALSE",
            "min_cov":3
        }
     ]
}
```

ML

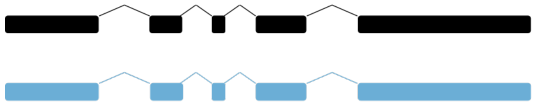
SQANTI variables

Transcript Attributes

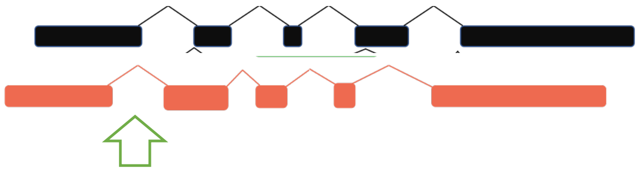
- Structural characteristics
  1. Detected/Reference length
  2. Distance to nearest annotated TSS
  3. Distance to nearest annotated TTS
- Coding potential
  1. ORF/CDS length
  2. CDS start and end position
- Quality control
  1. Number of indels
  2. PolyA intrapriming
  3. FSM class
- Support
  1. Number of Full-length reads.
  2. Minimum sample coverage
  3. Minimum coverage position
  1. Minimum splice junction coverage
- Expression levels:
  1. Transcript level
  2. Relative isoform abundance

True & Negative Set

True: FSM reference match



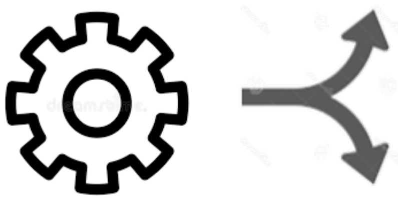
Negative: NNC – non canonical



Non-canonical

Random Forest

ALL transcripts

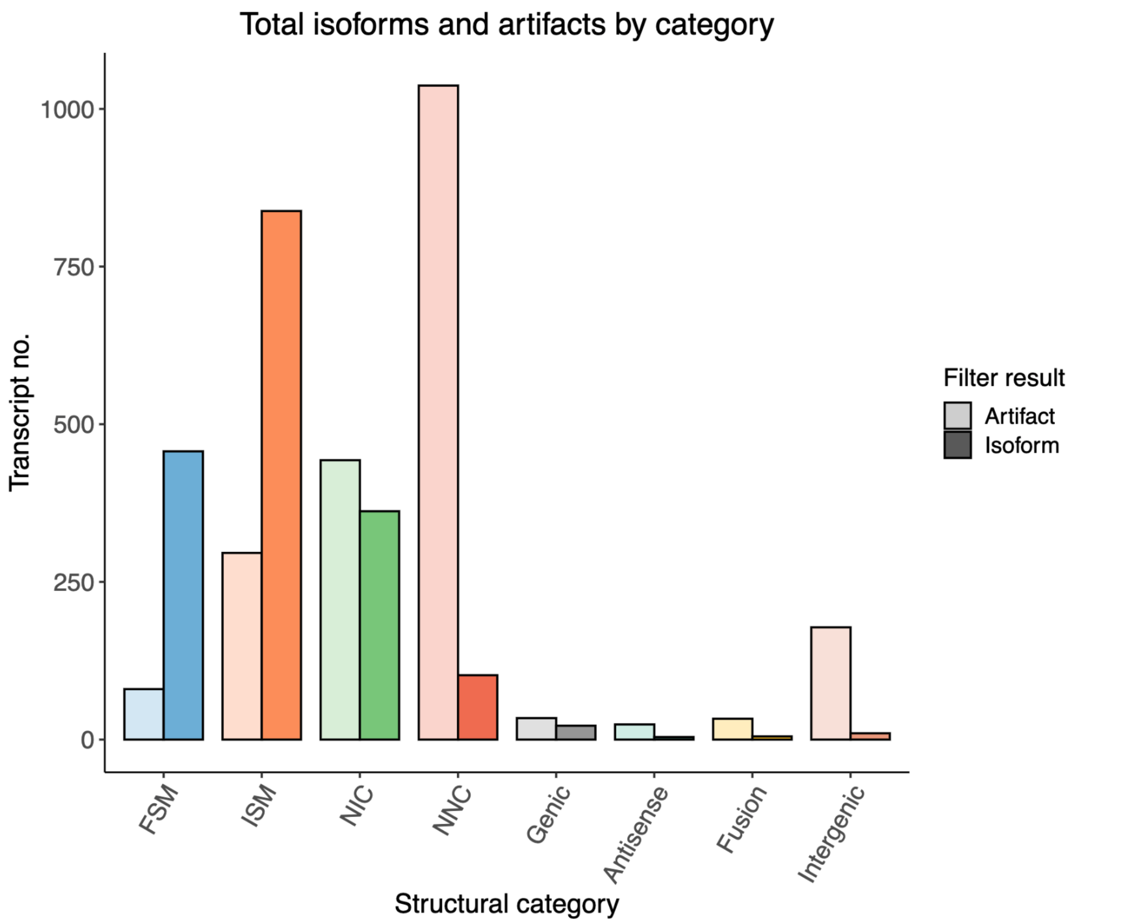


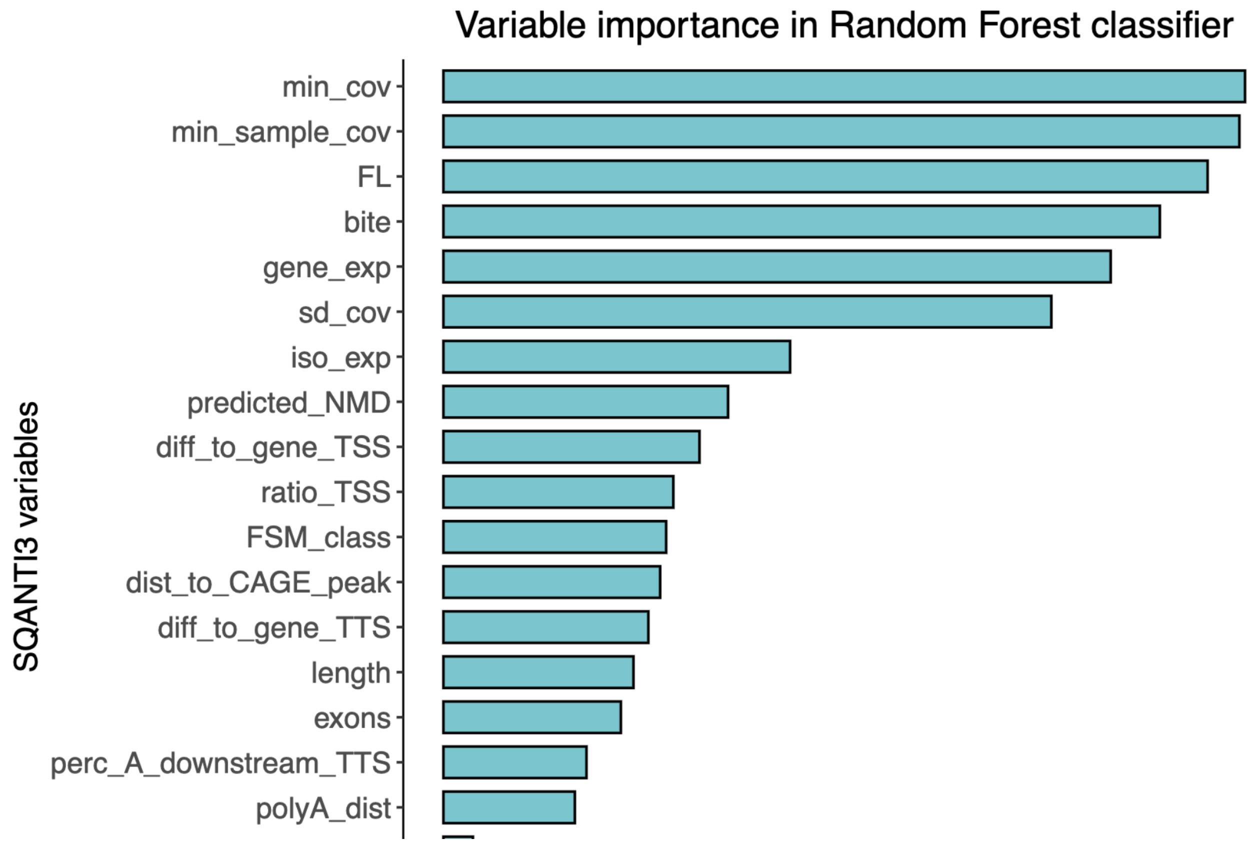
Isoforms

Artifacts

Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

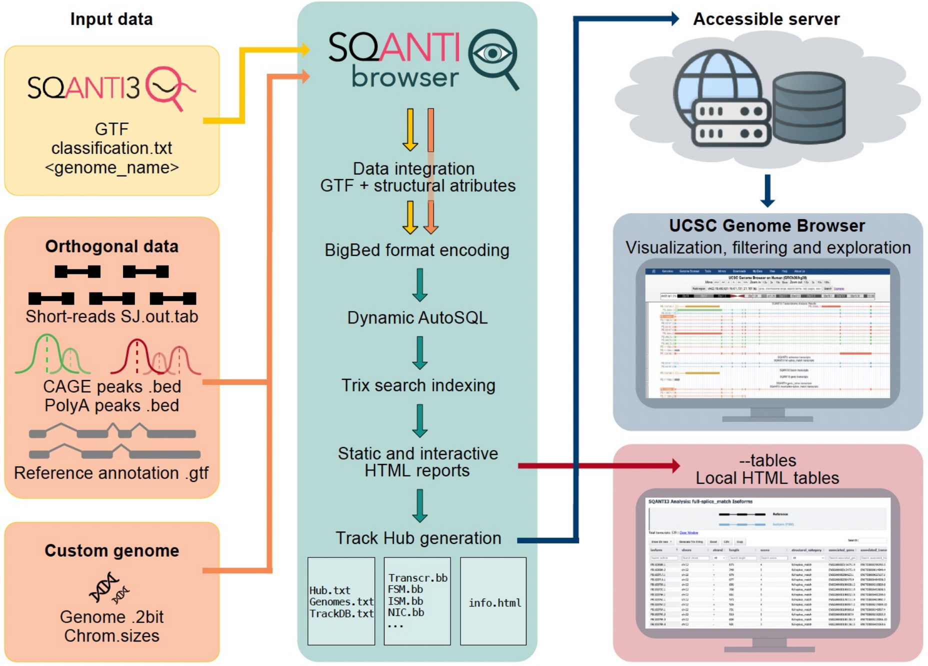
Clean transcriptome! And SQANTI-filter report





Source: Pardo-Palacios. 2024. 10.1038/s41592-024-02298-3

Bonus!! SQANTI-browser



Bonus!! SQANTI-verse

SQANTI3

SQANTI-filter

SQANTI-rescue

SQANTI-reads

SQANTI-tusco

SQANTI-single-cell

SQANTI-browser

SQANTI-proteins

IsoAnnot

TappAS

…

Ana Conesa
Fran Pardo-Palacios
Ángeles Arzalluz
Pablo Atienza
Alejandro Paniagua
Carlos Blanco
Andrés Colomer
Tianyuan Liu
Mark Diekhans
Fabian Jetzinger
Yalan Bi

All GGE members

Thank You!

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- https://github.com/ConesaLab/SQANTI-browser